



# **Dolby® Media Encoder User's Manual**

15 October 2015

Dolby Media Encoder Client v2.0

Dolby Media Encoder Server v2.0



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# Dolby Laboratories, Inc.

## Corporate Headquarters

**Dolby Laboratories, Inc.**  
100 Potrero Avenue  
San Francisco, CA 94103-4813 USA  
**Telephone** 415-558-0200  
**Fax** 415-863-1373  
[www.dolby.com](http://www.dolby.com)

## European Headquarters

**Dolby Laboratories, Inc.**  
Wootton Bassett  
Wiltshire SN4 8QJ England  
**Telephone** 44-1793-842100  
**Fax** 44-1793-842101

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# Introduction to the Dolby Media Encoder User's Manual

This manual describes how to use the Dolby® Media Encoder.

## 1.1 About This Documentation

The *Dolby Media Encoder User's Manual* provides content creators with workflows and supporting information for encoding from audio source files or a Dolby Atmos® master file set (home theater master) to a supported Dolby audio format in preparation for use with various targets (such as online media and Blu-ray Disc™).

The Dolby Media Encoder consists of two applications: Dolby Media Encoder Client and Dolby Media Encoder Server. Because all user-facing encoding controls are accessed through the client, most cases of Dolby Media Encoder in this manual refer to the client application, unless noted otherwise.

## 1.2 Dolby Media Encoder Software Application

Dolby Atmos Media Encoder lets you encode from audio source files or a Dolby Atmos master file set to a supported Dolby audio format in preparation for delivery on various media targets. Depending on the source material, the encoder supports encoding to a Dolby Digital, Dolby Digital Plus™, Dolby TrueHD, or MLP Lossless™ bitstream. Depending on the format, the encoder supports encodes targeted for Blu-ray Disc, streaming and downloadable media, DVD-Video, and DVD-Audio.

Dolby Media Encoder supports encoding from .wav audio source files to create a standard channel-based Dolby True HD, Dolby Digital Plus, Dolby Digital, and MLP Lossless bitstream, or encoding from a Dolby Atmos master file set to create a Dolby TrueHD or Dolby Digital Plus bitstream with Dolby Atmos, or a standard channel-based Dolby TrueHD, Dolby Digital Plus, or Dolby Digital bitstream as well.

The Dolby Media Encoder Client application accesses and sends jobs to a Dolby Media Encoder Server application, either on the same machine running the encoder, or a separate machine running the Dolby Media Encoder Server on a LAN. All user-facing encoding controls are available via the Dolby Media Encoder Client.

The Dolby Media Encoder Server application does the actual work of encoding. It runs as a background process on a local machine or a remote server, can be installed in multiple locations, and can be accessed by multiple clients.

### 1.3 Abbreviated Speaker Notations

The *Dolby Media Decoder User's Manual* documentation uses specific abbreviations for each speaker channel.

This table lists Dolby abbreviated speaker notations and corresponding speaker notations for other specifications. An em dash (—) indicates that no corresponding notation exists.

**Table 1-1** Abbreviated Speaker Notations

| Dolby Speaker Name             | Dolby Abbreviation | CEA-861 Abbreviation | ITU-R BS.2159-5 (Type B 10.2 Channel) Abbreviation | SMPTE 2036-2 (22.2 Channel) Abbreviation | IEC 62574 (30.2 Channel) Abbreviation |
|--------------------------------|--------------------|----------------------|----------------------------------------------------|------------------------------------------|---------------------------------------|
| Left                           | L                  | FL                   | L                                                  | FL                                       | FL                                    |
| Right                          | R                  | FR                   | R                                                  | FR                                       | FR                                    |
| Center                         | C                  | FC                   | C                                                  | FC                                       | FC                                    |
| Left Surround                  | Ls                 | RL                   | LS                                                 | —                                        | LS and LSd                            |
| Right Surround                 | Rs                 | RR                   | RS                                                 | —                                        | RS and RSd                            |
| Left Rear Surround/Left Back   | Lrs                | RLC                  | LB                                                 | BL                                       | BL                                    |
| Right Rear Surround/Right Back | Rrs                | RRC                  | RB                                                 | BR                                       | BR                                    |
| Center Surround                | Cs                 | RC                   | —                                                  | BC                                       | BC                                    |
| Left Wide                      | Lw                 | FLW                  | —                                                  | —                                        | FLW                                   |
| Right Wide                     | Rw                 | FRW                  | —                                                  | —                                        | FRW                                   |
| Left Center                    | Lc                 | FLC                  | —                                                  | FLC                                      | FLC                                   |
| Right Center                   | Rc                 | FRC                  | —                                                  | FRC                                      | FRC                                   |
| Left Surround 1                | Ls1                | —                    | —                                                  | SiL                                      | SiL                                   |
| Right Surround 1               | Rs1                | —                    | —                                                  | SiR                                      | SiR                                   |
| Left Front Height              | Lfh                | FLH                  | LH                                                 | TpFL                                     | TpFL                                  |
| Right Front Height             | Rfh                | FRH                  | RH                                                 | TpFR                                     | TpFR                                  |
| Left Top Middle                | Ltm                | —                    | —                                                  | —                                        | TpLS                                  |
| Right Top Middle               | Rtm                | —                    | —                                                  | —                                        | TpRS                                  |
| Left Top Front                 | Ltf                | —                    | —                                                  | TpSiL                                    | TpSiL                                 |
| Right Top Front                | Rtf                | —                    | —                                                  | TpSiR                                    | TpSiR                                 |
| Left Top Rear                  | Ltr                | —                    | —                                                  | —                                        | —                                     |
| Right Top Rear                 | Rtr                | —                    | —                                                  | —                                        | —                                     |
| Left Rear Height               | Lrh                | —                    | —                                                  | TpBL                                     | TpBL                                  |
| Right Rear Height              | Rrh                | —                    | —                                                  | TpBR                                     | TpBR                                  |
| Low-Frequency Effects 1        | LFE/LFE1           | LFE                  | LFE1                                               | LFE1                                     | LFE1                                  |
| Left Screen                    | Lsc                | —                    | —                                                  | —                                        | —                                     |
| Right Screen                   | Rsc                | —                    | —                                                  | —                                        | —                                     |
| Left Surround 2                | Ls2                | —                    | —                                                  | —                                        | —                                     |
| Right Surround 2               | Rs2                | —                    | —                                                  | —                                        | —                                     |
| Left Rear Surround 1           | Lrs1               | —                    | —                                                  | —                                        | —                                     |
| Right Rear Surround 1          | Rrs1               | —                    | —                                                  | —                                        | —                                     |
| Left Rear Surround 2           | Ls2                | —                    | —                                                  | —                                        | —                                     |
| Right rear Surround 2          | Rs2                | —                    | —                                                  | —                                        | —                                     |
| Left Center Surround           | Lcs                | —                    | —                                                  | —                                        | —                                     |
| Right Center Surround          | Rcs                | —                    | —                                                  | —                                        | —                                     |

**Table 1-1** Abbreviated Speaker Notations

| Dolby Speaker Name                | Dolby Abbreviation | CEA-861 Abbreviation | ITU-R BS.2159-5 (Type B 10.2 Channel) Abbreviation | SMPTE 2036-2 (22.2 Channel) Abbreviation | IEC 62574 (30.2 Channel) Abbreviation |
|-----------------------------------|--------------------|----------------------|----------------------------------------------------|------------------------------------------|---------------------------------------|
| Center Height                     | Ch                 | FCH                  | —                                                  | TpFC                                     | TpFC                                  |
| Top Surround                      | Ts                 | TC                   | —                                                  | TpC                                      | TpC                                   |
| Left Binaural                     | Lbin               | —                    | —                                                  | —                                        | —                                     |
| Right Binaural                    | Rbin               | —                    | —                                                  | —                                        | —                                     |
| Left Only                         | Lo                 | —                    | —                                                  | —                                        | —                                     |
| Right Only                        | Ro                 | —                    | —                                                  | —                                        | —                                     |
| Left Total                        | Lt                 | —                    | —                                                  | —                                        | —                                     |
| Right Total                       | Rt                 | —                    | —                                                  | —                                        | —                                     |
| Left Sound Direct                 | Lsd                | —                    | —                                                  | —                                        | —                                     |
| Right Sound Direct                | Rsd                | —                    | —                                                  | —                                        | —                                     |
| Left Dolby Enabled                | Le                 | —                    | —                                                  | —                                        | —                                     |
| Right Dolby Enabled               | Re                 | —                    | —                                                  | —                                        | —                                     |
| Left Surround Dolby Enabled       | Lse                | —                    | —                                                  | —                                        | —                                     |
| Right Surround Dolby Enabled      | Rse                | —                    | —                                                  | —                                        | —                                     |
| Left Rear Surround Dolby Enabled  | Lrse               | —                    | —                                                  | —                                        | —                                     |
| Right Rear Surround Dolby Enabled | Rrse               | —                    | —                                                  | —                                        | —                                     |

## 1.4 Contacting Dolby

You can contact Dolby regarding this product and its supporting documentation.

If you have technical questions about this product, please contact [dmpSupport@dolby.com](mailto:dmpSupport@dolby.com).

If you have questions or comments about this document, please contact [technical publications](#).

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# Installing and Launching Dolby Media Encoder

The Dolby® Media Encoder Client and Dolby Media Encoder Server are part of the Dolby Media Producer Suite. They are installed when the full suite is installed. Refer to the Dolby Media Producer Suite installation guide or what's new documentation for system requirements and installation steps.

## 2.1 Launching Dolby Media Encoder

The Dolby Media Encoder Client application provides access to all Dolby Media Encoder user-facing controls.

The Dolby Media Encoder Client application icon is labeled **Dolby Encoder Client**.

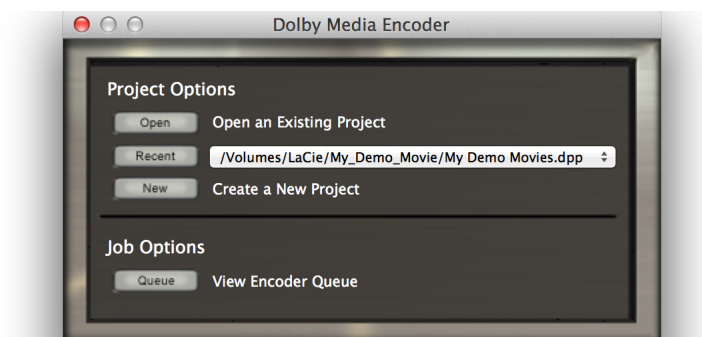


**Figure 2-1** Dolby Encoder Client Icon

To launch the Dolby Media Encoder, do one of the following:

- Click the Dolby Encoder Client application icon.
- Double-click the Dolby Encoder Client application file (located in /Applications/Dolby/Dolby Media Producer/).

On launch, the encoder displays the **Dolby Media Encoder** window.



**Figure 2-2** Dolby Media Encoder Window

The Dolby Media Encoder window provides access to some common project and job commands:

- **Project Options:**
  - **Open** button: Lets you browse for an existing project either locally or on the network. When you locate the project, click **Open**. The system then launches the job list window, where you can open an existing job in the project or add a new one.
  - **Recent** button and drop-down menu: Lets you select a recent project in the drop-down menu and then open it by clicking the button. By default, the most recent project is displayed in the menu.

After you click **Recent**, the system then launches the job list window, where you can open an existing job in the project or add a new one.

- **New** button: Opens the **Project Properties** window, where you can create a new project. After you create a project, you can then add jobs for encoding to it.
- **Job Options:**
  - **Queue** button: Opens the job list window, displaying the list of submitted encoding jobs.

## 2.2 Quitting Dolby Media Encoder

You can quit the Dolby Media Encoder session from the Dolby Media Encoder menu or icon.

To quit the application, do one of the following:

- In the Dolby Media Encoder menu, choose **Dolby Encoder Client > Quit Dolby Encoder Client**.
- Press <Command>+<Q>.
- Right-click the Dolby Encoder Client icon, and choose **Quit**.

## 2.3 Checking the Software Version Number

You can check the version number of the Dolby Media Encoder Client, including the version number of the parent Dolby Media Producer Suite.

To check the version number, do one of the following:

- Choose **Dolby Encoder Client > About Dolby Encoder Client** in the main window.
- Click the Dolby logo in the top-left corner of the job editor window (when available).



## Setting Up for Encoding with Remote Servers

Before you can do remote encoding with the Dolby® Media Encoder, you must configure the encoder for each remote server and mount file paths automatically or manually.

- [Adding a Remote Server](#)
- [Setting Up File Paths for Remote Servers](#)

### 3.1 Adding a Remote Server

In order to access and send encoding jobs to a Dolby Media Encoder Server application that is running on a LAN, you must add the server to the local client.

Before you begin:

- Make sure the client and server have their host name properly configured to connect the client to a new server.

To manually add a server to the client:

1. On the client machine, choose **Settings > Encoder Servers**.  
The system displays a dialog listing all servers added from your network.

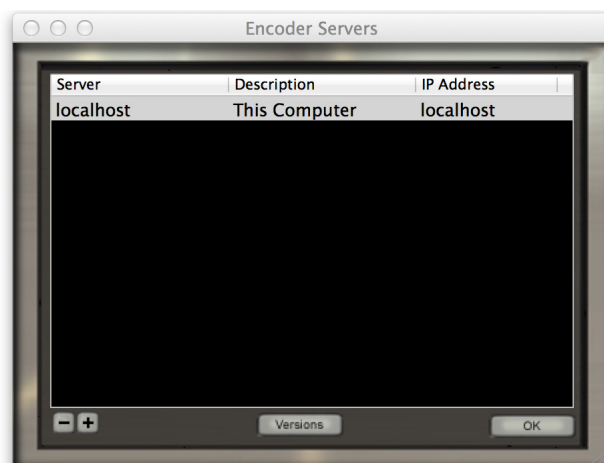


Figure 3-1 Encoder Servers Dialog

2. In the **Encoder Servers** dialog, click the **+** button at the lower left of the window to add a **Server** entry.  
The system creates a new line with three column fields, which respectively display the default name **new server**, the default description **remote encoder server**, and an empty **IP Address** field.

3. Enter the IP address or network name of the server (the machine where the Dolby Media Encoder Server program resides), by doing one of the following:
  - Double-click **new server** under the **Server** heading, and enter either the IP address or the network name of the server.
  - Right-click anywhere in the highlighted row, and then select from the list of available IP addresses and host names.

If you do not know the IP address of the Mac® computer with the installed server application, choose **System Preferences > Network** on that computer, and note the address in the **IP Address** field.

If you do not know the network name of the computer with the installed server application, choose **System Preferences > Sharing** on that computer. Note the network name in the line near the top of the window, labeled **Other computers on your local subnet can access your computer at**.

4. Optional: In the **Description** column, double-click the default description (**remote encoder server**) and type in a description.  
Use any text in the description; the system does not use this field.
5. Repeat steps 2 through 4 if you have additional servers for the client to use.
6. When done, click **OK** to close the **Encoders Servers** dialog.

### 3.1.1 Removing a Server

To remove servers from the client:

1. Select (highlight) the server in the list.
2. Click the – button at the lower left of the window.

### 3.1.2 Checking Server Versions

You can check the version number of the server.

To check a server version:

1. On the client machine, choose **Settings > Encoder Servers**.
2. Select (highlight) the server in the list.
3. Click the **Versions** button.

The **Server Version** dialog opens and displays the version number of the server.

## 3.2 Setting Up File Paths for Remote Servers

When encoding with a remote server, the encoder client needs to know both the path and location of source and destination files.

In many cases, you can set the paths and locations to be automounted (automatically detected and mounted). If automounting is not working, you must take action by manually mounting a drive, manually mapping file paths for the remote server, or both.

If some cases, you must also manually mount the server to the client, even if automounting will be enabled.

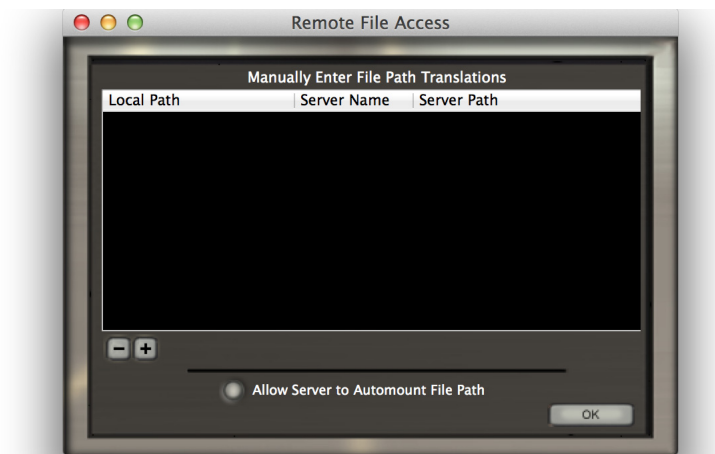
### 3.2.1 Automatically Mounting File Paths for a Remote Server

Generally, we recommend that you enable automatic detection and mounting of source and destination file paths.

If the source files are at the server, you must also manually mount the server to the client, even if automounting is enabled. See [Section 3.2.2](#).

To enable automatic detection and mounting:

1. Choose **Settings > Remote File Access**.  
The system opens the **Remote File Access** dialog.



**Figure 3-2** Remote File Access Dialog

2. When the dialog opens, click **Allow Server to Automount File Path**.
3. Click **OK** to close the dialog.

### 3.2.2 Manually Mounting Machines Using the Finder

If automounting does not work, you must mount source and/or destination locations using the Finder in order to be able to point to files from within the Dolby Media Encoder client. There are also scenarios where machines must be mounted even if automounting will be enabled.

Example scenarios where you must mount machines before enabling automounting include:

- Using a separate file server (see [Section 3.2.4](#)).
- Working with files stored on an external drive attached to the computer with the Dolby Media Encoder Server application installed (see [Section 3.2.5](#)).
- Working with files stored on an internal drive attached to the computer with the Dolby Media Encoder Server application installed (see [Section 3.2.6](#)).
- Working with files stored on an external drive attached to a client (see [Section 3.2.7](#)).

In the first three examples, mounting must be done from the client machine. In the last example, mounting must be done from the encoder server machine.

To manually mount directories that are not already mounted on a given Mac, connect to all the machines where servers and needed file directories are located using the **Connect to Server** option under the Finder **Go** menu (<Command>+<K>). This can be done by using the **Browse** button in the **Connect to Server** dialog, or by typing in the path to a shared directory directly in the **Server Address** text field at the top of the window. For additional information, refer to the documentation for your Mac OS®. To allow clients and servers to mount and share other storage devices, such as RAID drives, the server or computer needs to be configured to make these devices visible to all users.

### 3.2.3 Manually Mapping File Paths for a Remote Server

If automounting does not work, you may need to enter file path information manually.

If the mount point for a shared folder is different on the client and server, you must define a path translation entry.

Each mapped path in Dolby Media Encoder is defined by a file path entry (path translation entry). There is no limit to the number of file path entries that you can create for each server location.

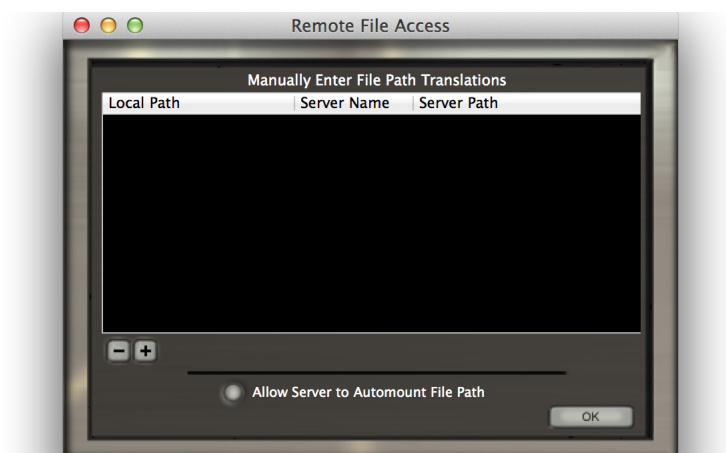
Example scenarios where you must manually map file paths for a remote server include:

- Files stored on an internal drive on the server
- Files stored on an external drive attached to a client

For more information on these examples, see [Section 3.2.4](#) through [Section 3.2.7](#).

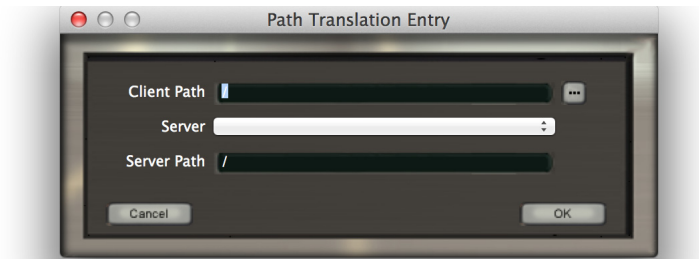
To map a file path:

1. Choose **Settings > Remote File Access**.  
The system opens the **Remote File Access** dialog.



**Figure 3-3** Remote File Access Dialog

2. Click the **+** button.  
The system displays the **Path Translation Entry** dialog.



**Figure 3-4** Path Translation Entry Dialog

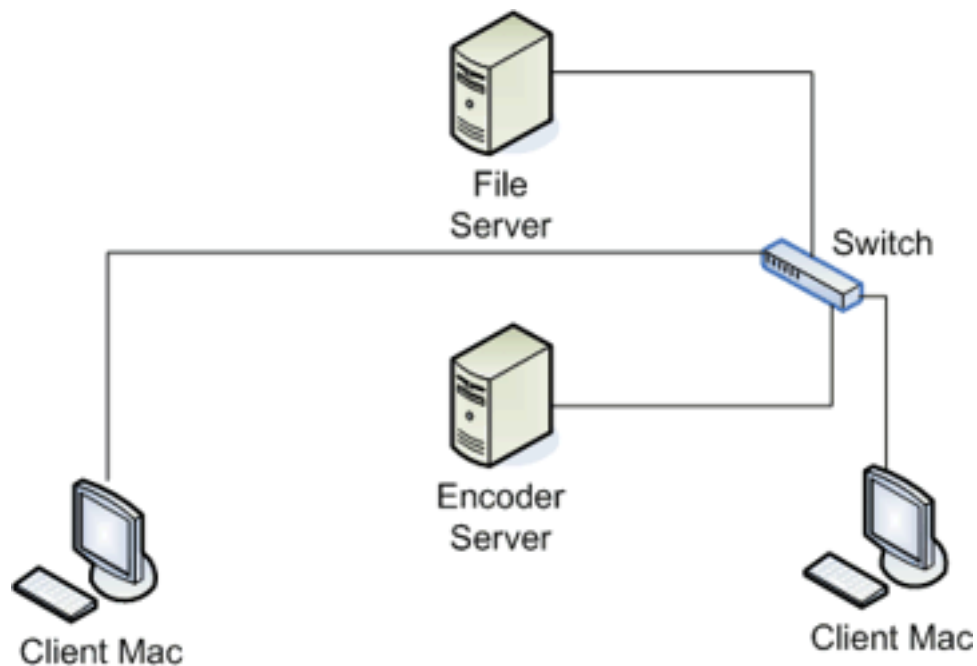
3. Enter a **Client Path** to a directory where either source files are stored or encoded files will be written. You can do this easily by clicking the browser button to the right of the **Client Path** field to open the Finder on the client computer and browse to the file location.
4. In the **Server** menu, choose the server for the intended paths.
5. Enter a **Server Path**.

The path entered here must correspond to the same path as entered in step 3, but as seen from the server. In most cases, this will use a slightly different path description to refer to the same data location.

If you are unsure of the exact path text to type here, you can discover the path by moving to the computer where the Dolby Media Encoder Server is installed, and using the Finder to browse to the same directory that you did for the client path. If you right-click the directory and choose **Get Info**, the server path to that directory appears on the **Where** line of the **Get Info** window.

6. When satisfied, click **OK** to save the new entry.
7. Repeat steps 2 through 6 for additional paths.

### 3.2.4 Scenario 1: Using a Separate File Server



**Figure 3-5** Network Scenario: Using a Separate File Server

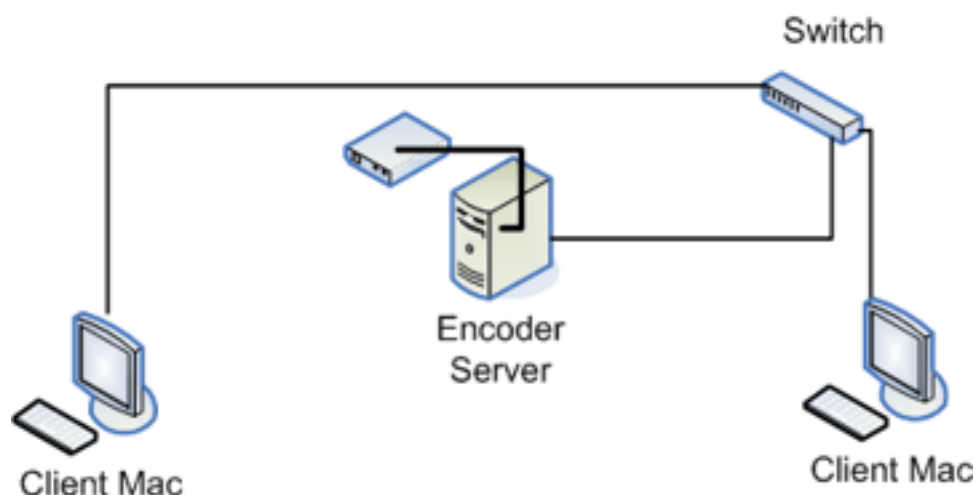
In this scenario, input files are stored on a machine that both the server and the client connect to over the network.

To set up file paths for the remote servers:

1. Manually mount the file server using the Finder on both the client and the encoder server.
2. Enable automounting.

You do not need to map any file paths because the path from the server to the files is the same as the path from the client to the files.

### 3.2.5 Scenario 2: Files Stored on an External Drive Attached to the Server



**Figure 3-6** Network Scenario: Files Stored on an External Drive Attached to the Server

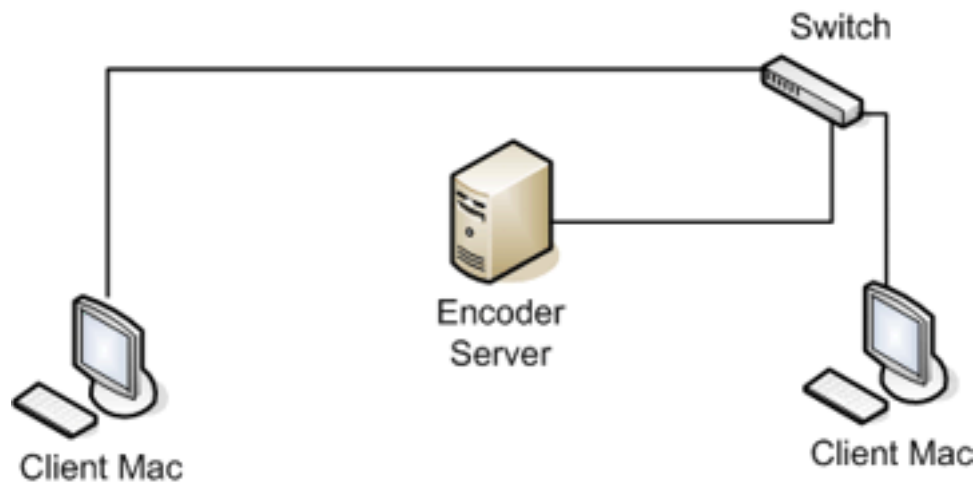
In this scenario, input files are on a separate drive physically attached to the machine where the server is located.

To set up file paths for the remote servers:

1. Ensure that the external drive on the server is designated as a shared drive in Mac OS X®.
2. Manually mount the encoder server drive and desired subdirectory from the client.
3. Enable automounting.

You do not need to map any file paths because the client machine sees the drive through the server.

### 3.2.6 Scenario 3: Files Stored on an Internal Drive on the Server



**Figure 3-7** Network Scenario: Files Stored on an Internal Drive on the Server

In this scenario, input files are stored on one of the server internal drives.

Automounting does not work in this scenario.

To set up file paths for the remote servers:

1. Manually mount the server internal drive on the client.
2. Map file paths, because the paths from the server and from the client are not the same.

A sample entry is:

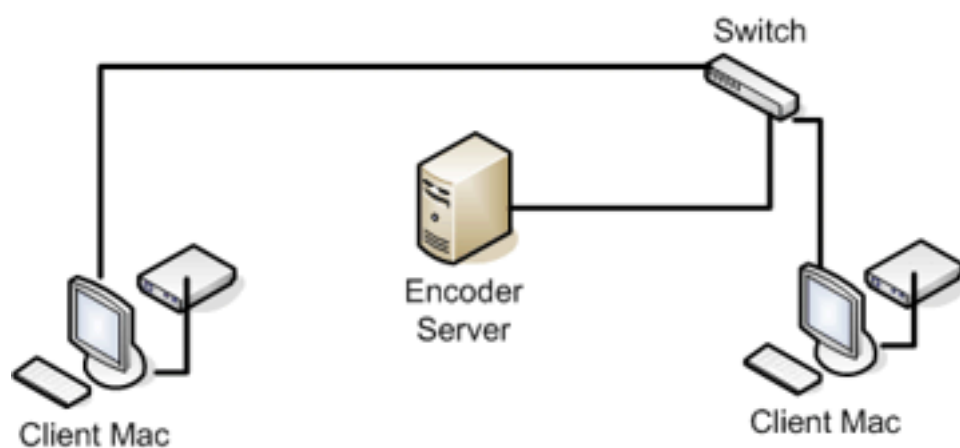
Local path: /Volumes/X-Serve/waves

Server name: 10.111.111.111

Server path: /Volumes/Files/G-Raid/Waves

In this example, G-Raid is the name of the internal drive.

### 3.2.7 Scenario 4: Files Stored on an External Drive Attached to a Client



**Figure 3-8** Network Scenario: Files Stored on an External Drive Attached to a Client

In this scenario, the encoder server application writes or reads encoded files to an external drive attached to a client machine.

To set up file paths for the remote servers:

1. Manually mount the external drive on the client to the server after it is attached to the client.  
This requires using the OS on the server machine to connect to the client hard drive over the network.
2. Enable automounting.
3. If automounting does not work, manually map file paths to allow the server to find the client drive.

A sample file-path translation entry for this situation is:

Local path: /Volumes/X-Serve/waves

Server name: 10.111.111.111

Server path: /Volumes/G-Raid/waves

In this example, G-Raid is the name of the external drive attached to the client machine.



# Encoding to a Dolby Audio Format

Use Dolby® Media Encoder to encode from audio source files or a Dolby Atmos® master file set to a supported Dolby audio format in preparation for delivery on various media targets.

## 4.1 Overview of Encoding to a Dolby Audio Format

The basic steps for encoding with Dolby Media Encoder are the same, regardless of the source material, the Dolby audio format, or the media targets.

The workflow for encoding to a Dolby audio format:

1. [Creating a New Project and a New Job](#)
2. [Choosing a Format and Media Target](#)
3. [Selecting Source Material \(Encoding Configurations\) for Encoding](#)
4. [Setting the 5.1 Downmix Type on the Channels Page \(Dolby Digital Plus and Dolby Digital Only\)](#)
5. [Loading Source Files](#)
6. [Specifying a Destination Path and File Name for an Encoded File](#)
7. [Setting Up the Encoder](#)
8. [Entering User Info for the Job](#)
9. [Submitting the Job for Encoding](#)

If you are unfamiliar with encoder jobs, see [Section 7.2](#).

For specific Dolby audio format workflows, see:

- [Creating a Dolby TrueHD Bitstream with or Without Dolby Atmos](#)
- [Creating a Dolby Digital Plus Bitstream with or without Dolby Atmos](#)
- [Creating a Dolby Digital Bitstream](#)

### 4.1.1 Creating a New Project and a New Job

Dolby Media Encoder requires the creation of a project as of version 2.0. This ensures that job records are stored correctly and allows retrieval of information for any job you create, even if not yet submitted for encoding.

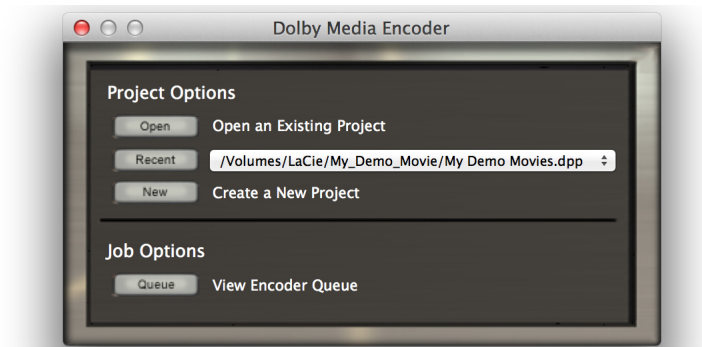
When you create a new project and save it, the application automatically creates the first a new job that you can configure for encoding. Alternatively, you can create new jobs in an existing project (see [Section 7.7](#)), or open an existing job in that project (see [Section 7.6.2](#)).

You can also create a Hot Job, which lets you run encoding jobs from a special folder that acts like an application. This feature provides semi-automated workflows for performing a series of similar jobs where the same parameters are desired for each job. See [Section 6.1](#) for more details.

To create a new project (and new job):

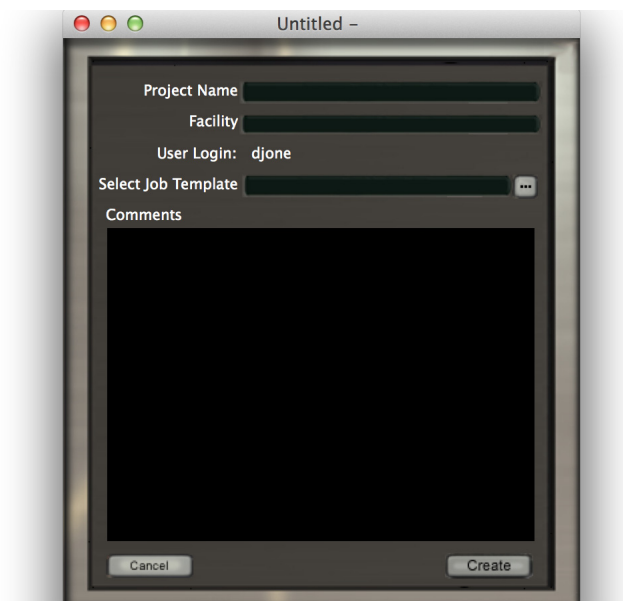
1. Launch the Dolby Media Encoder application.

When the application starts, it displays the **Dolby Media Encoder** startup window.



**Figure 4-1** Dolby Media Encoder Startup Window

2. In the **Project Options** section, click **New**.  
The **Project Properties** window opens.



**Figure 4-2** Project Properties Window

3. Enter information in the **Project Properties** window.  
New projects require a project name. All other fields are optional. Project properties can be modified later, if desired. For more information, see [Section 7.5](#).
4. Click **Create**.

5. When prompted in the **Save** dialog, do the following:
  - Save the project .dpp file with any name you choose at any desired location. The file name does not need to match the project name.
  - Click **OK**.

Once the project has been saved, the encoder opens the job list window and a job editor. You are now ready to configure the new job in the job editor.

#### 4.1.2 Choosing a Format and Media Target

For each job that you create, you must select a Dolby audio format to encode to and its media targets.

The encoding options available to you will vary depending on the format and targets chosen for the job. Choose them with care.

The format and target buttons are always displayed at the top of the job editor window, regardless of which job editor window page or subpage is displayed.

To choose a format and media targets:

1. In the job editor window, in the **Format** section, select a supported Dolby audio format:

- **Dolby Digital**
- **MLP**
- **Dolby Digital Plus**
- **Dolby TrueHD**

The default format is **Dolby TrueHD**.

2. In the Target section, select media targets.

The format you previously chose determines the range of targets available:

- If the format is **Dolby Digital**, the available targets are **DVD-Video** and **Blu-ray Disc**. You can choose either target individually, or both.
- If the format is **MLP**, the only available target is **DVD-Audio**.
- If the format is **Dolby Digital Plus**, the available targets are **Blu-ray Disc** and **Online Media**. You can choose one only.
- If the format is **Dolby TrueHD**, **Blu-ray Disc** and **Online Media** will be selected, because Dolby TrueHD streams formatted for Blu-ray Disc may also be streamed or downloaded.

You are now ready to load source files and specify a destination path for encoded files.

#### Multiple Targets and Encoding

When you have two media targets, the system produces a single encoded file that is compatible with both selected targets, instead of a separate file for each target. Attributes of these multiple-target files are restricted to the subset of format constraints common to both targets.

### 4.1.3 Selecting Source Material (Encoding Configurations) for Encoding

Encoding configurations for a job set the type of source files or downmixing that will be used to create an encoded file.

The format and targets of the job determine the available encoding configurations. See [Chapter 8](#). For the Dolby Digital Plus™ format, the Blu-ray Disc™ secondary audio status also determines which configurations will be available. (Choose the option for secondary audio encoding before selecting the encoding configuration.)

To select encoding configurations:

1. For Dolby TrueHD, do the following:
  - a. If you want to encode from a Dolby Atmos master file set, click (select) **Include Dolby Atmos**.
  - b. Click on the **Presentation** configuration drop-down menu for eight-, six-, or two-channel independent presentations, and select the desired channel format.

You may encode independent presentations from unique PCM source files for one, two, or three presentations, provided the program material complexity does not cause the data rate or file size to exceed your format limitations.

- c. For the remaining dependent presentations, click on the **Presentation** configuration drop-down menu for the desired presentation, and then select the appropriate downmix choice. For example, if independent PCM signals are provided for the eight-channel presentation, a dependent six-channel presentation will be created by downmixing the eight-channel independent presentation.
2. For Dolby Digital, Dolby Digital Plus, and MLP Lossless™ formats, click on the **Channel Config** drop-down menu and select the desired channel format.

If encoding to a Dolby Digital Plus bitstream with Dolby Atmos (**Dolby Atmos DD+ Bitstream** selected), also set the channel configuration of the backward-compatible presentation by clicking the **Audio Core Channel Configuration** drop-down menu and selecting one of the available configurations.

### 4.1.4 Setting the 5.1 Downmix Type on the Channels Page (Dolby Digital Plus and Dolby Digital Only)

When encoding to Dolby Digital Plus or Dolby Digital with 5.1 or more channels, you can typically select the type of downmix to be used for the 5.1 channel core of the encoded bitstream.

This option is also part of Dolby TrueHD job setup, but the option is on the **Downmix** subpage of the **Encoder Setup** page. Refer to [Section 9.1](#) for Dolby TrueHD encoding options.

The format and targets of the job determine whether 5.1 downmix settings are available on the **Channels** page.

To select the type of downmix to be used for the 5.1 core of the encoded bitstream:

- On the **Channels** page, click **5.1 Dolby Standard** (Lo, Ro) or **5.1 Dolby PLIIx**.  
When encoding from 7.1 PCM channels, the 5.1 core is made by downmixing the four rear channels to form two new ones. When encoding from an .atmos master, the 5.1 core is made by first rendering the Dolby Atmos audio (if loaded) to 7.1 and then

downmixing the four surround channels of the 7.1 into just two surround channels of the 5.1. This is the case whether the core channel configuration is 5.1 or 7.1. The options are **5.1 Dolby Standard** (Lo, Ro) and **5.1 Dolby PLIIx**.

The coefficients for a Dolby Pro Logic PLIIx downmix from 7.1 to 5.1 are:

$$L_s = L_{ss} + (-1.2 \text{ dB} \times L_{rs}) + (-6.2 \text{ dB} \times R_{rs})$$

$$R_s = R_{ss} + (-6.2 \text{ dB} \times L_{rs}) + (-1.2 \text{ dB} \times R_{rs})$$

The coefficients for a Standard (Lo, Ro) downmix from 7.1 to 5.1 are:

$$L_s = 0 \text{ dB} \times L_{ss} + 0 \text{ dB} \times L_{rs}$$

$$R_s = 0 \text{ dB} \times R_{ss} + 0 \text{ dB} \times R_{rs}$$

#### 4.1.5 Loading Source Files

You can load .wav .aif, or .atmos master files, depending on the format, target, and encoding configurations previously selected on the **Channels** page.

Dolby Media Encoder supports encoding of PCM source material (.wav or .aif) at different sample rates, dependent on the encoding format.

- Dolby Digital: Encodes at 48 kHz natively. Audio at 44.1, 88.2, 96, 176.4, and 192 kHz will be automatically sample-rate converted to 48 kHz.
- Dolby Digital Plus: Encodes at 48 kHz, natively. Audio at 44.1, 88.2, 96, 176.4, and 192 kHz will be automatically sample-rate converted to 48 kHz.
- MLP: Supports encoding at 44.1, 48, 88.2, 96, 176.4, and 192 kHz.
- Dolby TrueHD: Supports channel-based encoding at 44.1, 48, 96, and 192 kHz. 88.2 and 176.4 kHz encoding is not supported.



**Note:** The Blu-ray Disc specification does not support 44.1, 88.2, or 176.4 kHz.

To load source files:

1. Click on the browse button to the right of the audio field.
2. In the browser, locate the source file (.wav or .atmos master file, where applicable), highlight it, and click **Open** to add it.



**Tip:** The file selector can identify sets of properly named channel-based .wav source files. When you double-click the first file of a set of source files, the system selects the entire set automatically if it detects a complete set of files that follow the file-naming conventions detailed in [Section 11.1](#).

#### Clearing Files Loaded in the Encoder

If you make a mistake, click the **Clear** button to the left of the configuration identifier to erase the highlighted entry.

Holding <Ctrl> while clicking **Clear** clears all loaded source file entries for that configuration.

### 4.1.6 Specifying a Destination Path and File Name for an Encoded File

After you choose source files, the system fills in the destination path, encoded file name, and file extension automatically. You can change the default path and file name on the **Channels** page.

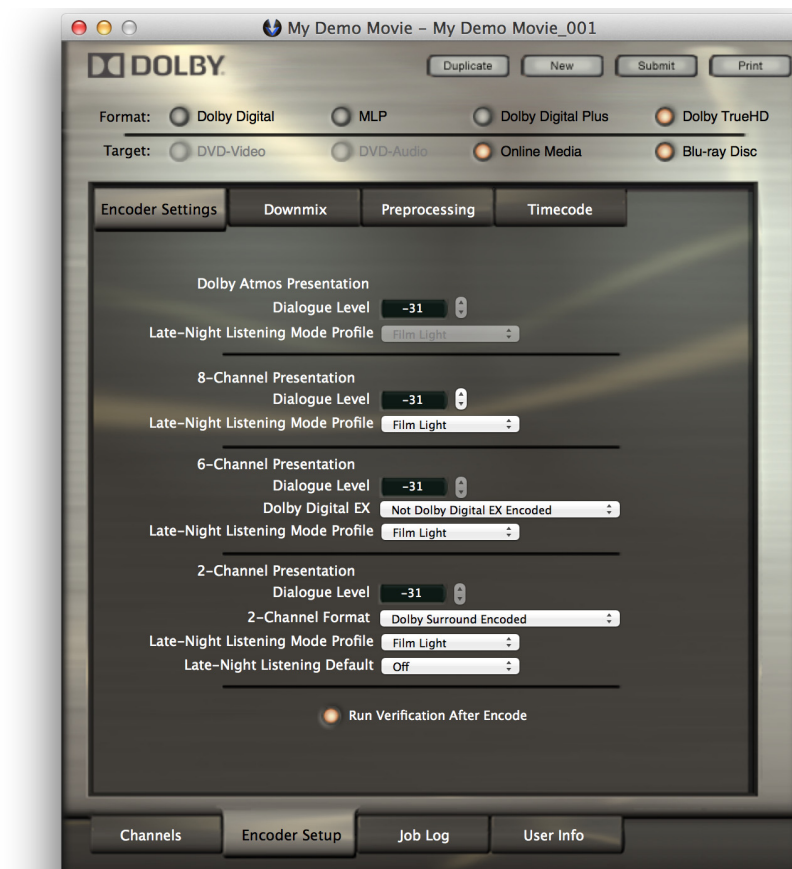
The destination path defaults to the source file directory. The file name defaults to the name of the first source file listed. The file extension is automatically set according to the selected format. The file extension updates dynamically if the format is changed.

To specify a new destination path and file name, do one of the following:

- Use the browse button:
  - a. Click on the browse button to the right of the channel field.
  - b. Optional: Browse to a new location.
  - c. Optional: Type in a new name.
  - d. Click **Save**.
- In the **Destination Path** field, type or paste in the path and file name.

### 4.1.7 Setting Up the Encoder

Dolby Media Encoder automatically inserts industry-standard default values for each encoder setting and metadata parameter required for the encoding process, based on the selected output format. You can make changes to these settings on the **Encoder Setup** subpages.



**Figure 4-3** Example of the Encoder Setup Page with Encoder Settings Subpage Displayed

There are four subpages accessible from the **Encoder Setup** page:

**Encoder Settings:** This is the first and most important **Encoder Setup** subpage. Use this subpage to adjust the settings for dialogue level, data rate, dynamic range control, and other parameters. The exact options available are determined by the selected format.

**Downmix:** Use this subpage to adjust various downmix settings. The exact options available are determined by the selected format.

**Preprocessing:** Use this subpage to adjust the settings for filtering, surround attenuation, and other parameters. The exact options available are determined by the selected format. For Dolby TrueHD, this page is where you will find the option to apply advanced upsampling (to 88.2 or 96 kHz).

**Timecode:** Use this subpage to enable or disable timecode embedding, as well as to choose the timecode format and starting time for the embedded code. You can also specify a timecode range to encode when encoding just a portion of the input file, and you can choose to prepend silence to the beginning of the file in milliseconds or frames.

For specific steps on aligning audio and video, see [Section 4.3](#).

A different set of encoder options is available for each Dolby audio format

- [Dolby Digital Encoding Options](#)
- [Dolby Digital Plus Encoding Options](#)
- [Dolby TrueHD Encoding Options](#)
- [MLP Lossless Encoding Options](#)

#### 4.1.8 Entering User Info for the Job

You can provide user information for each job. This includes the job name, facility, and comments.

By default, the encoder client automatically populates the first two fields on the **User Info** page. The job name defaults to a sequence number added to the project name, the facility is carried over from the previous facility name, or the facility name entered for the project, and the login is based on the computer user name.

You can keep or change the job name and the facility entry. If you accept the system defaults, you do not need to update the information on this page. Use the **Comments** section to enter information for later reference.



**Figure 4-4** User Info Page

To update user information for the job:

1. In the job editor window, click the **User Info** button.
2. Optional: Type in new information in the **Job Name** and **Facility** fields.
3. Optional: Add comments in the **Comments** section.
4. Choose **File > Save**.

#### 4.1.9 Submitting the Job for Encoding

To encode files, you submit jobs to the server application on a local or remote computer for encoding.



**Note:** You cannot encode files with a remote server until you have properly [configured your network](#).



Before you begin:

- If submitting a job to a remote server:
  - Make sure that the client and server volume names are different.
  - If two drives are attached to the client, where one drive is the source and the other drive is the destination, mount both on the server, and then make sure both of those are entered appropriately into the **Path Translation Entry** dialog.

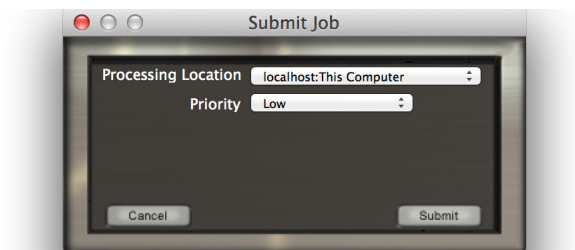
To submit a job for encoding:

1. Click the **Submit** button at the top right of the job editor.



**Figure 4-5** Submit Button

- If the project has not been saved, you will be prompted to create a project document for the job. Click **Yes**. The system displays the **Submit Job** dialog.
- If the project has been saved, the system displays the **Submit Job** dialog.



**Figure 4-6** Submit Job Dialog

2. Select a processing location from the drop-down menu, which provides a list of your available (configured) servers. If only one server is installed, that server will be the default location. If multiple servers are installed, the client will default to the last used location.
3. Select a priority from the **Priority** drop-down menu. The priority that is selected affects the place of the job in the encoding queue. Higher priority jobs are encoded first.
4. Click **Submit** to begin the process.

If your network requires authentication, the system will display up to two login dialogs.

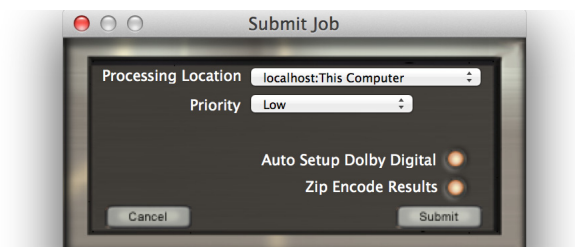
The **Audio Source Auto-Mount Certification** requires a login name and password for access to the computer storing the source files. The **Audio Destination Auto-Mount Certification** requires a login name and password for the destination machine. If the source and destination are on the same machine, the second dialog does not appear.

## Dolby TrueHD for Blu-ray Disc Encoding and Dolby Digital Soundtrack Requirement

The Blu-ray Disc specification requires a Dolby TrueHD job to have an accompanying Dolby Digital soundtrack (with a data rate of up to 640 kbps). These two encodes are multiplexed in authoring and presented to the consumer as a single soundtrack choice. The

home listener's player automatically selects the highest-quality version that it supports. The Blu-ray Disc specification refers to this combined track as Dolby Lossless.

To simplify your workflow, the **Submit Job** dialog has an additional option, **Auto Setup Dolby Digital**.

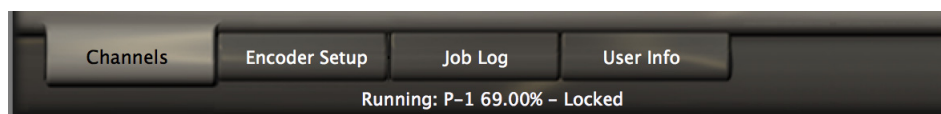


**Figure 4-7** Dolby TrueHD Submit Dialog

When the **Auto Setup Dolby Digital** button is enabled, the encoder automatically launches a Dolby Digital job editor with all of the alike settings copied over from your Dolby TrueHD job. The default data rate is 640 kbps.

## 4.2 Monitoring Encoding Progress

The system reports encoding progress as a percentage complete at the bottom of the job editor. This percentage can also be seen in the job list window queue list.



**Figure 4-8** Progress Report

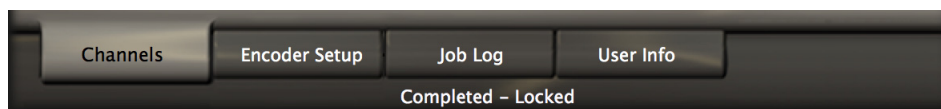
Because Dolby TrueHD encoding is a multipass process that can take time, Dolby Media Encoder provides additional information about the pass state. Dolby Atmos jobs have two additional pass states, for spatial coding and matrixing.

**Table 4-1** Dolby TrueHD Encoding Passes

| Encoder Actions                                                                                                      |
|----------------------------------------------------------------------------------------------------------------------|
| Spatial coding (Dolby Atmos jobs only)                                                                               |
| Matrixing (Dolby Atmos jobs only)                                                                                    |
| Analyze; this pass may repeat several times.                                                                         |
| Encode.                                                                                                              |
| Minimize data rate.                                                                                                  |
| Verify (optional, selected by clicking <b>Run Verification After Encode</b> on the <b>Encoder Settings</b> subpage). |

After a job has been submitted, its attributes can no longer be changed. The job editor displays the status **Locked**.

When encoding finishes, the status changes to **Completed—Locked**.



**Figure 4-9** Locked Status

Encoded jobs are automatically locked to ensure that you always have an accurate record of your encode. A locked job can always be duplicated if you wish to reuse its settings.

## 4.3 Ensuring that Encoded Audio Aligns with Video

Prior to encoding, you may specify the time range for encoding, to tell the encoder to encode just a specific portion of the input file. This is done to ensure that the encoded audio aligns with the encoded video.

When mastering for Dolby Atmos, we recommend adding audio preroll (greater than four seconds) to the audio in Pro Tools®, prior to recording the master file. Then, after importing the .atmos master into Dolby Media Encoder, you set the audio encoding start point to align with the video start frame (the first video frame for encoding, sometimes called the “first frame of action” or the “first frame of picture”). You can set the encode range relative to the source start timecode that is embedded in the input, or as a “file position” that you specify (with the start of the file treated as time 00:00:00.000).

### 4.3.1 Specifying the Time Range to Encode, Relative to Embedded Timecode

Prior to encoding, you may specify the time range for encoding, relative to the source start timecode that is embedded in the input.



**Figure 4-10** Specifying the Time Range to Encode, Related to Embedded Timecode

To specify the time range to encode, relative to embedded timecode:

1. Click the **Encoder Setup** tab, and then the **Timecode** tab to open the **Timecode** subpage.
2. Click (highlight) the **Specify Time Range to Encode** button.
3. In the **Time Range Relative to** drop-down menu, select **Embedded Timecode**.

If this option is used for files without embedded timecode, a starting timecode of 00:00:00:00 will be used.

4. In the **Encode Start** time range drop-down menu, select the desired fps or millisecond (ms) option.
5. For Dolby Atmos material, note the **Dolby Atmos Master Offset Time** displayed at the top of the window.

This is the start time of the input content, which you will use in the next step.

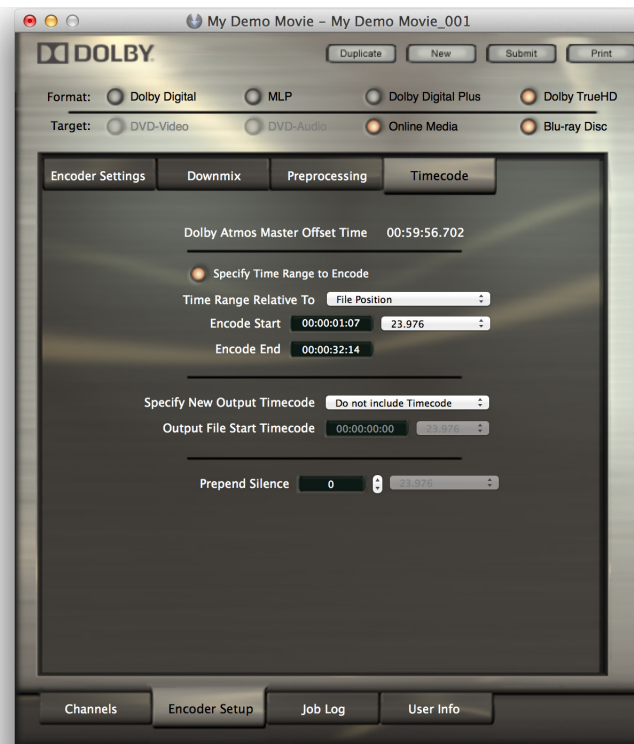
6. In the **Encode Start** field, enter the timecode at which you wish to begin encoding. Usually, this time is the same as the desired start of encoded video (first frame of action), for example, 0:59:58:00. This time must be the same or later than the time noted in the previous step.
7. Optional: Enter a different **Encode End** time, which is before the source end time. The **Encode End** field is automatically set to the very end of the input.

Change the time only if you wish to trim a portion from the end of your input file.

8. Optional: On the **Timecode** subpage, specify the output file timecode frame rate and start time, as well as prepend silence to the encoded file.

### 4.3.2 Specifying the Time Range to Encode, Relative to File Position

Prior to encoding, you may specify the time range for encoding as a specified “file position.”



**Figure 4-11** Specifying the Time Range to Encode, Related to File Position

To specify time range to encode, relative to file position:

1. Click the **Encoder Setup** tab, and then the **Timecode** tab to open the **Timecode** subpage.
2. Click (highlight) the **Specify Time Range to Encode** button.
3. In the **Time Range Relative** to drop-down menu, select **File Position**.
4. In the **Encode Start** time range drop-down menu, select the desired fps or millisecond (ms) option.
5. In the **Encode Start** field, enter the timecode at which you wish to begin encoding.

Usually this time should be the same as the desired start of the video (often called the first frame of action). This start point is interpreted relative to the start of the input file, which is treated as time "zero" (ignoring any time stamps in the input) when using **File Position**.

6. Optional: Enter a different **Encode End** time, which is before the source end time. The **Encode End** field is automatically set to the very end of the input.

Change the time only if you wish to trim a portion from the end of your input file.

7. Optional: On the **Timecode** subpage, specify the output file timecode frame rate and start time, as well as prepend silence to the encoded file.

## 4.4 Creating a Dolby TrueHD Bitstream with or Without Dolby Atmos

You can encode from a Dolby Atmos master file set (home theater master), to create a Dolby TrueHD bitstream with or without Dolby Atmos metadata.

When encoding from a home theater master to Dolby TrueHD, and including one or more independent channel-based presentations from .wav files in the same job, the **Encode Start** and **Encode End** times on the **Timecode** subpage define the encoded duration, as well as the start- and end-time points of the time range within the Dolby Atmos master source that will be encoded. However, for the independent presentations from .wav inputs, the **Encode Start** and **Encode End** times define the duration only. The encoding will start from the beginning of the original .wav files. To ensure the encoding captures the desired .wav audio data, trim the beginning of the .wav files in a digital audio workstation prior to importing them into Dolby Media Encoder.



**Note:** When encoding legacy channel-based presentations only, without including Dolby Atmos, the **Encode Start** and **Encode End** times on the **Timecode** subpage will define both the duration and the sample-accurate start time of the .wav files.

To create a Dolby TrueHD bitstream with or without Dolby Atmos:

1. Launch Dolby Media Encoder.
2. Create and save a project to create a new job, or open an existing project and add a new job to that project.
3. In the **Format** section, select **Dolby TrueHD**.

In the **Target** section, the system automatically selects **Online Media** and **Blu-ray Disc**. The system will produce a single encoded file that is compatible with both selected targets. Attributes of these multiple-target files are restricted to the subset of format constraints common to both targets.

4. Select the source material for encoding:
  - a. If you want to encode from a Dolby Atmos master file set, click (select) **Include Dolby Atmos**.



**Figure 4-12** Job Editor Window Set Up to Encode to Dolby TrueHD with Dolby Atmos

- b. Click on the **Presentation** configuration drop-down menu for eight-, six-, or two-channel independent presentations, and select the desired channel format.

You may encode independent presentations from unique PCM source files for one, two, or three presentations, provided the program material complexity does not cause the data rate or file size to exceed your format limitations. See [Section 8.1](#).

To encode from a Dolby Atmos master file set to a Dolby TrueHD bitstream with Dolby Atmos, click on the **8-Channel Presentation** drop-down menu and select **Downmix from Dolby Atmos**. To encode from a Dolby Atmos master file set to a channel-based 7.1 Dolby TrueHD bitstream (without Dolby Atmos), select **Dolby Atmos render to 7.1 (Lrs, Rrs)**. The default is **Downmix from Dolby Atmos**.

- c. For the remaining dependent presentations, click on the **Presentation** configuration drop-down menu for the desired presentation, and then select the appropriate downmix choice. For example, if independent PCM signals are provided for the eight-channel presentation, a dependent six-channel presentation will be created by downmixing the eight-channel independent presentation.



**Note:** The **Copy of** options for creating dependent presentations are not supported when encoding from a Dolby Atmos master file set to Dolby TrueHD. **Copy of** options are still supported for creating dependent presentations for encodes to Dolby TrueHD that do not include Dolby Atmos material.



5. If encoding from a Dolby Atmos Master file set, load the master file by doing the following:
  - a. Click on the browse button to the right of the **Include Dolby Atmos** field.
  - b. In the browser, locate the .atmos master file, highlight it, and click **Open** to add it to the presentation.

This step effectively adds all three master source files to the encoder.

For information about the loaded Dolby master file set, click the **Details** button. The following information is provided: presentation type, timecode information, encryption, Dialnorm, PCM file, metadata file, bed channel list, total bed channels, and total objects.



**Figure 4-13** Dolby Atmos Master File Set Loaded for Encoding to Dolby TrueHD with Dolby Atmos

6. Add PCM source files, as needed, for each channel-based presentation.
  - Click on the browse button to the right of the channel field.
  - In the browser, locate the .wav source file, highlight it, and click **Open** to add it.



**Tip:** The file selector can identify sets of properly named source files. When you double-click the first file of a set of source files, the system selects the entire set automatically if it detects a complete set of files that follow the file-naming conventions detailed in [Section 11.1](#).



**Note:** For eight-, six-, or two-channel independent presentations, at least one input channel needs to be loaded. The encoder will pad unloaded channels with silence.



**Note:** When encoding with both an .atmos master and PCM source signals to include multiple independent presentations in one Dolby TrueHD stream, the timecode settings will apply only to the .atmos master source. Make sure the PCM source signals are already trimmed to the desired start and end time points before loading them.



**Figure 4-14** Example of Dolby Atmos and PCM Source Files Loaded for Encoding to Dolby TrueHD

7. Verify the destination path in the **Destination Path** field.

After you choose source files, the system fills in the destination path, encoded file name, and file extension automatically. The destination path defaults to the source file directory. The file name defaults to the name of the first source file listed. The file extension is automatically set according to the selected format. The file extension updates dynamically if the format is changed.

You can change these values by typing new names or browsing to a new location.

8. Set up the Encoder.

Dolby Media Encoder automatically inserts industry-standard default values for each encoder setting and metadata parameter required for the encoding process, based on the selected output format. You can make changes to these settings on the **Encoder Setup** subpages: **Encoder Settings**, **Downmix**, **Preprocessing**, and **Timecode**. See [Section 9.1](#).

For specific steps on aligning audio and video on the **Timecode** subpage, see [Section 4.3](#).



When encoding from a Dolby Atmos master file set, there are options specific to the master that you need to configure:

**Number of Elements** setting, which represents the number of clustered signals of Dolby Atmos material to be output by spatial coding. Set the **Number of Elements**, based on the following:

- **16:** Use this setting if you have no data space considerations and want the absolute best quality.
- **14:** Use this setting if space allows it and an extra measure of quality is desired.
- **12:** Use this setting for Blu-ray Discs with space considerations and excellent sound quality. **12** is the default number of elements, because it provides high quality at an efficient average and peak data rate.

Leave the **Bed Configuration** and **Apply Magnetic Centroid** options at their default setting, unless Dolby support recommends different settings.

Use the **Optimize Data Rate** option to apply more aggressive quantization to intermediate signals used in the encoder signal path when encoding from a Dolby Atmos master file set to Dolby TrueHD. This can reduce the peak and average data rates by 10-15% for typical content.



**Figure 4-15** Preprocessing Subpage for Encoding to Dolby TrueHD

9. Enter user info for the job.

The Comments field is an optional area for notes or comments about the job. For more information, see [Section 4.1.8](#).

10. Submit the job.

When prompted, select whether to automatically create a Dolby Digital bitstream and whether to bundle the Dolby TrueHD bitstream, the Dolby Digital bitstream and the Dolby TrueHD encoding log file into a .zip archive after the encoding is done.

For more information, see [Section 4.1.9](#).

When encoding finishes, the status changes to **Completed-Locked**. An .mlp file is now available at the designated destination.

## 4.5 Creating a Dolby Digital Plus Bitstream with or without Dolby Atmos

You can encode from a Dolby Atmos Master file set (home theater master), to create a Dolby Digital Plus bitstream with or without Dolby Atmos audio and associated metadata.

To create a Dolby Digital Plus bitstream with or without Dolby Atmos:

1. Launch Dolby Media Encoder.
2. Create and save a project to create a new job, or open an existing project and add a new job to that project.
3. Select a format and target:
  - a. In the **Format** section, select **Dolby Digital Plus**.
  - b. In the **Target** section, select **Online Media** or **Blu-ray Disc**.
4. Select a channel configuration:
  - a. Click on the **Channel Config** drop-down menu and select the desired channel format.

To encode from a Dolby Atmos master file set to a Dolby Digital Plus bitstream with Dolby Atmos, select **Downmix Atmos DD+ Bitstream**. To encode from a Dolby Atmos master file set to a 7.1 or 5.1 channel Dolby Digital Plus bitstream (without Dolby Atmos), select **Dolby Atmos render to 7.1 (Lrs, Rrs)** or **Dolby Atmos render to 5.1 (Lrs, Rrs)** (online media only).

See [Section 8.2](#).

- b. If encoding to a Dolby Digital Plus bitstream with Dolby Atmos (**Downmix Atmos DD+ Bitstream** selected), set the channel configuration of the backward-compatible presentation by clicking the **Audio Core Channel Configuration** drop-down menu and selecting one of the available configurations.

This menu is not available if you selected **Dolby Atmos render to 7.1 (Lrs, Rrs)** in the previous step (for example, to encode from a Dolby Atmos master file set to a Dolby Digital Plus bitstream without Dolby Atmos).

5. Load source files.

For each file, do the following:

- a. Click on the browse button to the right of the audio field.
  - b. In the browser, locate the respective source file (.atmos master file or .wav), highlight it, and click **Open** to add it.

When loading an .atmos master file, this step effectively adds all three master source files to the encoder.



**Tip:** The file selector can identify sets of properly named channel-based .wav source files. When you double-click the first file of a set of source files, the system selects the entire set automatically if it detects a complete set of files that follow the file-naming conventions detailed in [Section 11.1](#).

---



**Note:** The presentation must have at least one input channel loaded. When a channel is not loaded, the encoder pads it with silence.

6. Verify the destination path in the **Destination Path** field.

After you choose source files, the system fills in the destination path, encoded file name, and file extension automatically. The destination path defaults to the source file directory. The file name defaults to the name of the first source file listed. The file extension is automatically set according to the selected format. The file extension updates dynamically if the format is changed.

You can change these values by typing new names or browsing to a new location.

7. Select the type of downmix to be used for the 5.1 backward-compatible core of the Dolby Digital Plus bitstream.

The 5.1 core is made by first rendering the Dolby Atmos audio (if loaded) to 7.1 and then downmixing the four surround channels of the 7.1 into just two surround channels of the 5.1. This is the case whether the core channel configuration is 5.1 or 7.1. The options are **5.1 Dolby Standard** (Lo, Ro) and **5.1 Dolby PLIIx**.

The coefficients for a Dolby Pro Logic PLIIx downmix from 7.1 to 5.1 are:

$$\begin{aligned} L_s &= L_{ss} + (-1.2 \text{ dB} \times L_{rs}) + (-6.2 \text{ dB} \times R_{rs}) \\ R_s &= R_{ss} + (-6.2 \text{ dB} \times L_{rs}) + (-1.2 \text{ dB} \times R_{rs}) \end{aligned}$$

The coefficients for a Standard (Lo, Ro) downmix from 7.1 to 5.1 are:

$$\begin{aligned} L_s &= 0 \text{ dB} \times L_{ss} + 0 \text{ dB} \times L_{rs} \\ R_s &= 0 \text{ dB} \times R_{ss} + 0 \text{ dB} \times R_{rs} \end{aligned}$$

8. Set up the Encoder.

Dolby Media Encoder automatically inserts industry-standard default values for each encoder setting and metadata parameter required for the encoding process, based on the selected output format. You can make changes to these settings on the **Encoder Setup** subpages: **Encoder Settings**, **Downmix**, **Preprocessing**, and **Timecode**. See [Section 9.2](#).

For specific steps on aligning audio and video on the **Timecode** subpage, see [Section 4.3](#).

When encoding from a Dolby Atmos master file set, there are options specific to the master that you need to configure:

**Number of Elements** setting, which represents the number of clustered signals of Dolby Atmos material to be output by spatial coding. Set the **Number of Elements** to 12 for all Dolby Digital Plus jobs:

- **12** is the default number of elements, and provides the optimal balance between object metadata and perceptual coding data.

Leave the **Bed Configuration** and **Apply Magnetic Centroid** options at their default setting, unless Dolby support recommends different settings.

9. Enter user info for the job.

For more information, see [Section 4.1.8](#).

10. Submit the job.

For more information, see [Section 4.1.9](#).

When encoding finishes, the status changes to **Completed—Locked**. An .ec3 file (for online media) or .eb3 file (for Blu-ray Disc) is now available at the designated destination.

## 4.6 Creating a Dolby Digital Bitstream

You can encode audio from a 5.1 channel rendering of a Dolby Atmos Master file set (home theater master) or standard channel-based audio .wav files to create a Dolby Digital bitstream.

When encoding from a Dolby Atmos master file set, the bitstream will not carry Dolby Atmos metadata. However, it will sound the same as the 5.1 channel presentation heard when playing a Dolby Digital Plus with Dolby Atmos metadata or Dolby TrueHD bitstream with Dolby Atmos metadata. This bitstream is typically generated for use on Blu-ray Discs, which require a backward-compatible Dolby Digital track to accompany Dolby TrueHD.

To create a Dolby Digital bitstream:

1. Launch Dolby Media Encoder.
2. Create and save a project to create a new job, or open an existing project and add a new job to that project.
3. Select a format and target:
  - a. In the **Format** section, select **Dolby Digital**.
  - b. In the **Target** section, select **DVD-Video**, **Blu-ray Disc**, or both.
4. Select a channel configuration:
  - Click on the **Channel Config** drop-down menu and select the desired channel format. To encode from a Dolby Atmos master file set, select **Dolby Atmos render to 5.1**. See [Section 8.3](#).
5. Load source files.

For each file, do the following:

  - a. Click on the browse button to the right of the audio field.
  - b. In the browser, locate the respective source file (.atmos master file or .wav), highlight it, and click **Open** to add it.

When loading an .atmos master file, this step effectively adds all three master source files to the encoder.



---

**Tip:** The file selector can identify sets of properly-named channel-based .wav source files. When you double-click the first file of a set of source files, the system selects the entire set automatically if it detects a complete set of files that follow the file naming conventions detailed in [Section 11.1](#).

---



---

**Note:** The presentation must have at least one input channel loaded. When a channel is not loaded, the encoder pads it with silence.

---

6. Verify the destination path in the **Destination Path** field.

After you choose source files, the system fills in the destination path, encoded file name, and file extension automatically. The destination path defaults to the source file directory. The file name defaults to the name of the first source file listed. The file extension is automatically set according to the selected format. The file extension updates dynamically if the format is changed.

You can change these values by typing new names or browsing to a new location.

7. Select the type of downmix to be used for the 5.1 backward-compatible core of the Dolby Digital bitstream.

The 5.1 core is made by first rendering the Dolby Atmos audio (if loaded) to 7.1 and then downmixing the four surround channels of the 7.1 into just two surround channels of the 5.1. This is the case whether the core channel configuration is 5.1 or 7.1. The options are **5.1 Dolby Standard** (Lo, Ro) and **5.1 Dolby PLIIx**.

The coefficients for a Dolby Pro Logic PLIIx downmix from 7.1 to 5.1 are:

$$Ls = Lss + (-1.2 \text{ dB} \times Lrs) + (-6.2 \text{ dB} \times Rrs)$$

$$Rs = Rss + (-6.2 \text{ dB} \times Lrs) + (-1.2 \text{ dB} \times Rrs)$$

The coefficients for a Standard (Lo, Ro) downmix from 7.1 to 5.1 are:

$$Ls = 0 \text{ dB} \times Lss + 0 \text{ dB} \times Lrs$$

$$Rs = 0 \text{ dB} \times Rss + 0 \text{ dB} \times Rrs$$

8. Set up the Encoder.

Dolby Media Encoder automatically inserts industry-standard default values for each encoder setting and metadata parameter required for the encoding process, based on the selected output format. You can make changes to these settings on the **Encoder Setup** subpages: **Encoder Settings**, **Downmix**, **Preprocessing**, and **Timecode**. See [Section 9.3](#).

For specific steps on aligning audio and video on the **Timecode** subpage, see [Section 4.3](#).

9. Enter user info for the job.

For more information, see [Section 4.1.8](#).

10. Submit the job.

For more information, see [Section 4.1.9](#).

When encoding finishes, the status changes to **Completed—Locked**. An .ac3 file is now available at the designated destination.

## 4.7 Creating an MLP Lossless Bitstream

You can encode audio from standard channel-based audio .wav files to create an MLP Lossless bitstream.

To create an MLP Lossless bitstream:

1. Launch Dolby Media Encoder.
2. Create and save a project to create a new job, or open an existing project and add a new job to that project.

3. In the **Format** section, select **MLP**.  
The MLP format supports the **DVD-Audio** target only.
4. Select a channel configuration:
  - Click on the **Channel Config** drop-down menu and select the desired channel format. See [Section 8.4](#).
5. Load source files.  
For each file, do the following:
  - a. Click on the browse button to the right of the audio field.
  - b. In the browser, locate the respective source file (.atmos master file or .wav), highlight it, and click **Open** to add it.



---

**Tip:** The file selector can identify sets of properly-named channel-based .wav source files. When you double-click the first file of a set of source files, the system selects the entire set automatically if it detects a complete set of files that follow the file naming conventions detailed in [Section 11.1](#).

---



---

**Note:** The presentation must have at least one input channel loaded. When a channel is not loaded, the encoder pads it with silence.

---

6. Verify the destination path in the **Destination Path** field.  
After you choose source files, the system fills in the destination path, encoded file name, and file extension automatically. The destination path defaults to the source file directory. The file name defaults to the name of the first source file listed. The file extension is automatically set according to the selected format. The file extension updates dynamically if the format is changed.

You can change these values by typing new names or browsing to a new location.

7. In the **2-Channel Downmix** drop-down menu, select whether to use a downmix of the 6-channel for the stereo backward-compatible core of the MLP bitstream, or to not to include a downmix.
8. Set up the Encoder.  
Dolby Media Encoder automatically inserts industry-standard default values for each encoder setting and metadata parameter required for the encoding process, based on the selected output format. You can make changes to these settings on the **Encoder Setup** subpages: **Encoder Settings**, **Downmix**, **Preprocessing**, and **Timecode**. See [Section 9.4](#).
9. Enter user info for the job.  
For more information, see [Section 4.1.8](#).
10. Submit the job.  
For more information, see [Section 4.1.9](#).

When encoding finishes, the status changes to **Completed—Locked**. An .mlp file is now available at the designated destination.

## Managing Submitted Jobs in the Encoder Queue

You can view and manage submitted jobs in the encoder queue of the job list window. Additionally, you can view submitted jobs remotely on a Dolby® Media Encoder web page from a browser.

### 5.1 Viewing Active Jobs in the Encoder Queue

You can see all the jobs that are currently encoding, queued, or completed by the selected server. You can pause jobs, move them up or down in the queue, and remove them. You can also open a submitted job and duplicate it for use in a new encode.

To view active jobs in the encoder queue:

1. In the job list, click **Encoder Queue**.
2. Click **Active**.

### 5.2 Viewing Archived Jobs in the Encoder Queue

You can see all of the encoding jobs that have been completed and autoarchived.

To view archived jobs in the encoder queue:

1. In the job list, click **Encoder Queue**.
2. Click **Archived**.

### 5.3 Sorting Jobs in the Encoder Queue

You can sort the order of jobs in the encoder queue by sorting them.

By default, the job list displays jobs in the encoder queue in the following order:

1. Jobs currently being encoded.
2. Jobs submitted for encoding, sorted by their place in the line
3. Jobs completed, sorted by completion time. The most recently completed jobs are listed at the bottom.



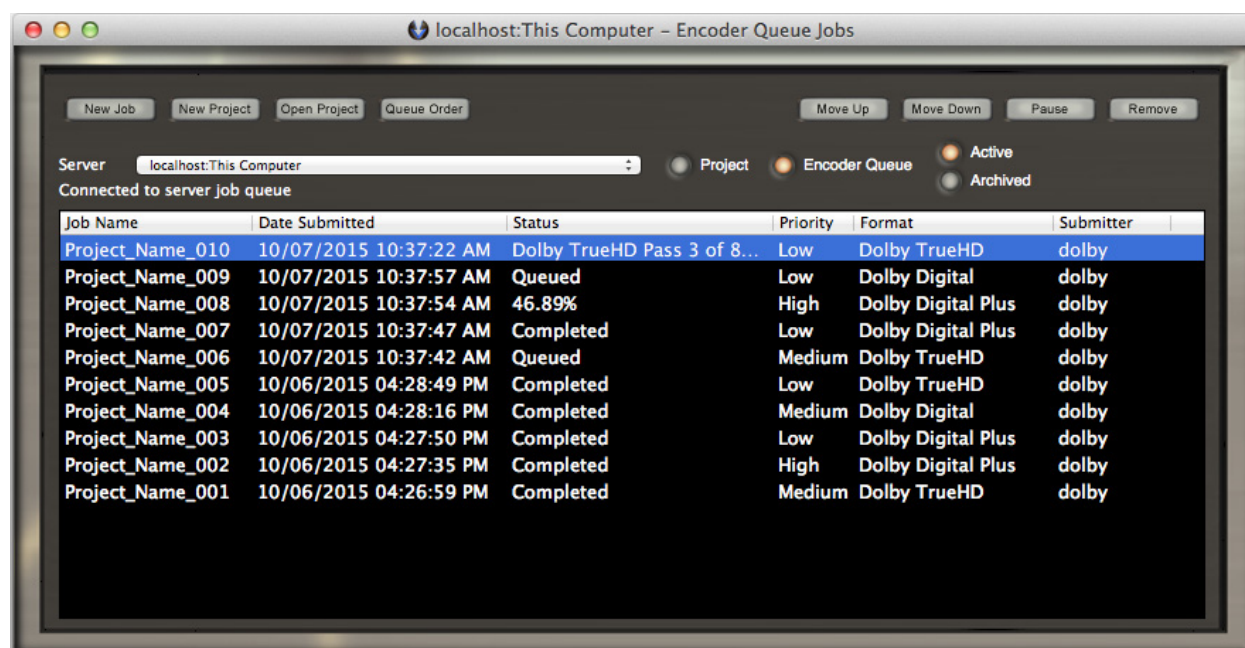


Figure 5-1 Job List Encoder Queue

To change the order of jobs in the job list:

- Click the heading of any column to sort by that value.
- Click **Queue Order** to return the page to the default sort order.
- Move jobs up and down in the list using the **Move Up** and **Move Down** buttons. See the next section.

### 5.3.1 Managing Jobs in the Encoder Queue

You can move, pause, resume, or remove jobs in the encoder queue by using action buttons or right-click commands.

Only jobs that are not yet encoded can be moved, paused, or resumed. Any job can be removed. When a job is paused, the **Pause** button changes to **Resume**. Click **Resume** to continue encoding.



Figure 5-2 Encoder Queue Action Buttons



To move, pause, resume, or remove jobs:

1. In the job list, click **Encoder Queue**.
2. Do one of the following:
  - Right-click on a job and select an action.
  - Select one or more jobs in the list, and then click the desired action button located at the upper right of the job list window.

The encoder supports selecting one job by clicking (highlighting) it, or selecting multiple jobs by clicking (highlighting) a job and then selecting additional jobs by using the <Shift> or <Command> key.

### Action Buttons for Jobs in the Encoder Queue

Dolby Media Encoder provides action buttons to manage one or more jobs in the encoder queue.

Action buttons include:

- **Move Up:** Moves the highlighted jobs up one position in the queue
- **Move down:** Moves the highlighted jobs down one position in the queue
- **Pause (and Resume):** Pauses and resumes processing of the highlighted jobs
- **Remove:** Removes the highlighted job from the queues

### Right-Click Menus for Jobs in the Encoder Queue

Each job in the encoder queue includes right-click menu commands to manage the job.

Menu commands include:

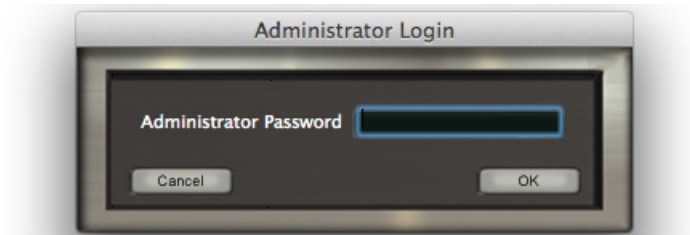
- **Move Up:** Moves the highlighted job up one position in the queue
- **Move down:** Moves the highlighted job down one position in the queue
- **Move top:** Moves the highlighted job to the top of its priority group
- **Move bottom:** Moves the highlighted job to the bottom of its priority group
- **Pause and Resume:** Pauses and resumes processing of the highlighted job
- **Open:** Opens the job editor window for the highlighted job
- **Remove:** Removes the highlighted job from the queue

## 5.3.2 Managing Jobs Submitted on the Client or By Someone Else

You can manage jobs submitted on the local client, or by someone else on a different client.

While viewing the active encoder queue, you are always allowed to pause, resume, or remove your own jobs (that is, those submitted by your client).

If you attempt to change a job submitted by another user, the system requires you to enter the administrative password before it will make the change. The system displays the **Administrator Login** dialog.



**Figure 5-3** Administrator Login Dialog



---

**Note:** The password is case sensitive. The default is *Dolby*. We recommend that you change it upon installing the application.

---

## 5.4 Displaying Submitted Jobs in Your Browser

You can use your browser to display submitted jobs and their job logs.

The browser window displays the server name, an activity summary, a list of all jobs in order of priority, and their status.

Before you begin, you must enable web services:

1. Navigate to Applications/Utilities.
2. Launch Terminal.
3. Type `sudo apachectl start` in the command line.
4. Press <Enter>.

To display submitted jobs in a browser:

- In your browser, type in the IP address or network name of your server, followed by the directory path `/cgi-bin/Dolby/DolbyCGI`. For example:
  - `http://10.115.101.110/cgi-bin/Dolby/DolbyCGI`
  - `http://fzb2.local/cgi-bin/Dolby/DolbyCGI`



---

**Note:** This page is optimized for Apple® Safari®. Its functions have not all been tested

---

### 5.4.1 Job List Source

In the drop-down menu, you can choose **Active**, which shows jobs queued, jobs encoding, or jobs recently completed; or **Archived**, which shows completed jobs that have already been autoarchived.

## Advanced Features

The Dolby® Media Encoder provides a comprehensive set of tools to make encoding more efficient and to store relevant job setup information. The following sections describe these tools:

- [Encoding Files in a Hot Job Folder](#)
- [Creating Copies of Jobs for Reuse](#)
- [Reading and Saving the Log](#)
- [Printing Job Editor Information](#)

### 6.1 Encoding Files in a Hot Job Folder

A Hot Job is a folder that acts like an application. You can drag and drop sets of source files into it. The folder then automatically sets up and queues jobs using the chosen settings. The folder can be used to automatically set up as many separate encoding jobs as you like.



**Note:** Hot job encoding does not support encoding to a remote server or encoding to Dolby TrueHD or Dolby Digital with Dolby Atmos® bitstreams.

Hot Jobs created with previous versions of Dolby Media Encoder are not compatible with Dolby Media Encoder version 2.0, and will need to be created again.

To encode files in a Hot Job folder:

1. Do one of the following:
  - Choose **Job > New Hot Job**.
  - With a Hot Job open, choose **Job > Duplicate**.

A job editor window opens, with the **Submit** button now labeled **Save**.

The Hot Job folder name can be edited in the **Job Name** field in the **User Info** page of the job. The default is HotJob.app, but this can be changed. The source file section is grayed out because the source files you use will be the ones dropped into the folder.

2. Choose a destination path. This will be the location for your encoded jobs.
3. Choose settings on the **Encoder Setup** pages that will apply to all the jobs generated by the folder.
4. Click the **Save** button.

The folder for the Hot Job appears on the desktop.

5. To encode a job, simply drag and drop the source files into the Hot Job folder.



**Note:** The encoded file name defaults to the name of the first file, normally Left. The files you drop in must conform to the file-naming convention discussed in Section 11.1.

If the number of files or naming convention do not match the Hot Job setting, you will get a rejected error message.

Multiple jobs created using identically named source files are identified in the encoded file name by a number appended to the source file name: with a Left input file of myjob.L.wav, first job name is myjob.\_1.ac3, with subsequent jobs named myjob.\_2.ac3, myjob.\_3.ac3, and so on.

If you create a separate Hot Job that uses the same destination folder, and then run a job with the same input files used with a previous job, the original, existing output files will be written by the new Hot Job (without any alert).

The files you drop are not physically moved or copied into the Hot Jobs folder. They are only pointed to.

You can open, review, and change settings in the Hot Job folder at any time by double-clicking the folder itself. Clicking the **Save** button when updating is completed saves the updates and closes your Hot Job.

## 6.2 Creating Copies of Jobs for Reuse

The duplication and template features of the Dolby Media Encoder allow you to make copies of the settings for a job on the fly and to save them separately for reuse.

### 6.2.1 Duplicating a Job

Job duplicates are useful when you want to reuse the settings of an existing job (or Hot Job) to update segments of an existing job. Making a duplicate is also the only way to change the settings of a job that is completed.

To duplicate a job (or Hot Job):

1. Open the job (or Hot Job) you wish to duplicate in the job editor.
2. Do one of the following:
  - Choose the **Duplicate Job** command in the job editor.
  - Choose **Job > Duplicate Job**.

The system opens a job with identical settings and **\_copy** appended to its name. All settings are the same, except the destination path will be empty.

### 6.2.2 Saving a Job as a Template

Saving common job settings as a template saves you time by eliminating the need to re-create a job step by step.

To save a template:

1. Open the job (completed or new) you wish to save as a template.
2. Choose **Job > Save as Template**.

The system opens a standard **Save** dialog.

If you have files loaded into the job, their file names and paths will be included in the saved template.

3. Enter a name for the template, and choose a destination for it.

By default, the **Save** dialog shows the contents of `//Users/<username>/Library/Application Support/Dolby/`. We recommend that templates be stored in this directory so they are easily located by the client in the future. You can also save them in another directory of your choice.

4. Click **Save**.

The system saves the file with a `.djt` extension.

### 6.2.3 Working with Job Templates

Once saved, a template can be used in two ways:

- You can apply it to an entire project by entering it in the **Select Job Template** field in the **Project Properties** window. When a template is selected, new jobs opened in that project will use that template instead of the application default.
- You can also load it for an individual job by choosing **Job > Load Template**.

Choosing **Load Template** lets you select a saved template from the folder of your choice. Once you load a template, a new job editor will open with the settings for the template.

Choosing **Delete Template** opens the template folder and allows you to select a template to delete.

## 6.3 Reading and Saving the Log

The encoder automatically creates a log for each job and generates it in the **Job Log** page. The log lists all successful step completions and any error conditions that occur.

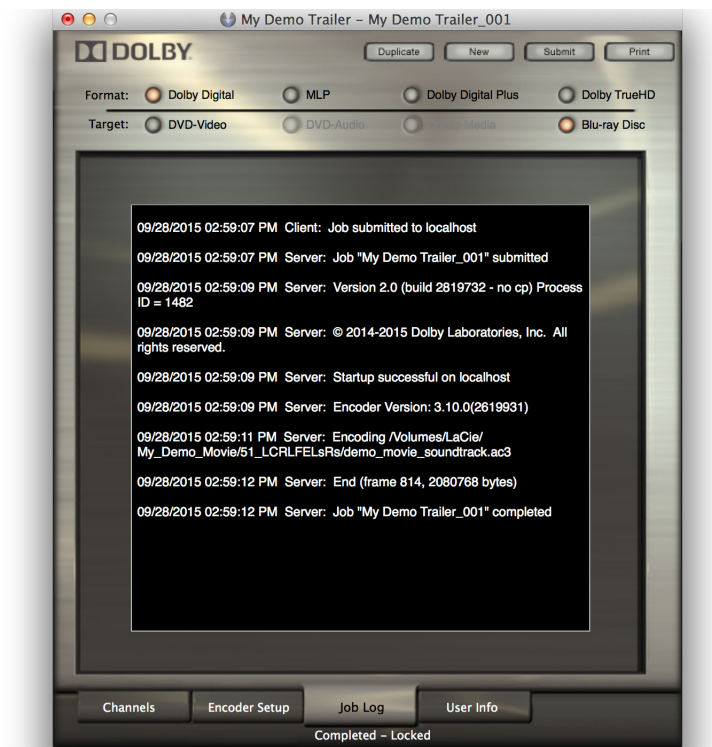


Figure 6-1 Job Log Page

## 6.4 Printing Job Editor Information

You can print a completed list of all your job information at any time. The document may be sent to a printer or saved as a .pdf file. Page 1 lists all of the format and target information, as well as the source files used and the destination path. Page 2 lists all of the settings: data rate, metadata, and so on. Page 3 lists the job log.

To print a job:

1. Do one of the following:
  - From the menu bar, choose **File > Print Job Info**.
  - In the job, click the **Print** button.
2. Set the parameters and start the print job.

## 6.5 Troubleshooting

### 6.5.1 Encoding Problems

The first step in troubleshooting an encoding problem is viewing the job log. Duplicate the job to create a new one, correct any error conditions noted in the previous job log, and resubmit the job.

## 6.5.2 Trouble Connecting to the Server

If you receive a **Cannot connect to server** error, follow these steps:

1. Verify the name of the server.  
If the client and the server are running on the same machine, the server name must end in .local.
2. Make adjustments using the **Encoder Servers** dialog, accessible via the **Settings** menu.
3. Verify that an iLok™ USB key is inserted into the computer where the Dolby Media Encoder Server is installed.
4. Verify that your network is properly setup.



**Note:** Machines connected to dynamic IP networks are sometimes assigned new IP addresses. If your client cannot find the server, it can be helpful to open the **Encoder Server** window and right-click the server in the list. If nothing appears in the right-click drop-down menu, this indicates that the server or machine either is not functioning correctly or is not connected to the network.

If you see the server, reselect it and try again. Reselecting will update the server IP address.

If you do not know your IP address, you can find the address for any machine in the **System Preferences > Network** menu.

---

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# Managing Projects and Jobs

The Dolby® Media Encoder uses projects to organize encodes and job information.

## 7.1 Project Definition

A project file contains all of the encoding information for all of the jobs that were created in that project, including jobs that were created but not yet submitted. Much like you would use a folder to hold and organize related documents on your computer, a project file is used to manage related encoding jobs.

## 7.2 Job Definition

In this application, a job is the information used to create an encoded file, not the encoded file itself. Job files are kept in the project file in which they were created. A job is created by launching a job editor to select source files, the destination of your encoded file, metadata settings, and much more. When you finish selecting settings for your encoding job, submit it to the server to be queued for encoding. Once you submit a job, you are free to create a new job or resume work on other encoding jobs while the submitted job is encoded.

Completed jobs can be duplicated to save time when creating a new job with the same settings. Jobs can also be accessed later, so you can review the settings and source files that were used and where the encoded file was stored. You can keep detailed records of all the job encoding done in your facility for future reference.

## 7.3 Using a Project

A project can be used many different ways to help you manage your encoding jobs and make it easy to find previously encoded job edits. This includes letting you review job logs, encoder settings, and files used.

Here are some common ways that you can use a project:

1. Create a project for a particular movie. Then encode all your jobs for that movie in that project. All of the job encoding information for that movie is kept together.
2. Create a project file for a particular customer. Then encode all your jobs for that customer in that project. All of the job encoding information for that customer is kept together.
3. Create a project for each day or week in your facility. Then encode all your jobs for that day or week in that project. All of the job encoding information for that day or week is kept together.
4. Create a separate project for each user. All of the job encoding information for each user is kept separate.

5. Create a separate project for each encode you do. That project file will contain all of the information for that one encode.

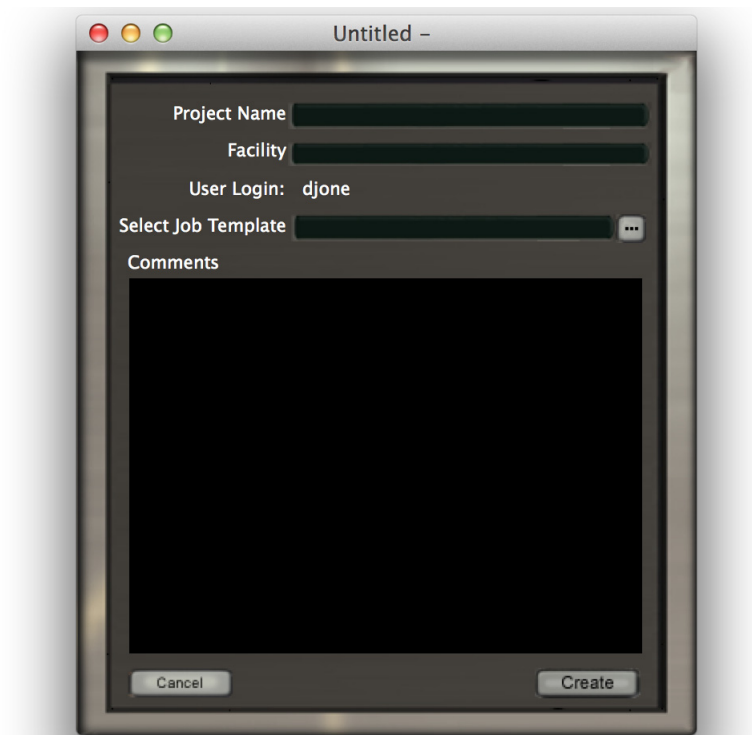
## 7.4 Creating a New Project

When you create a new project and save it, you automatically create a new job that you can configure for encoding.

To create a project:

1. Launch the Dolby Media Encoder application.
2. In the **Project** section, click **New Project**.

The **Project Properties** window opens.



**Figure 7-1** Project Properties Window

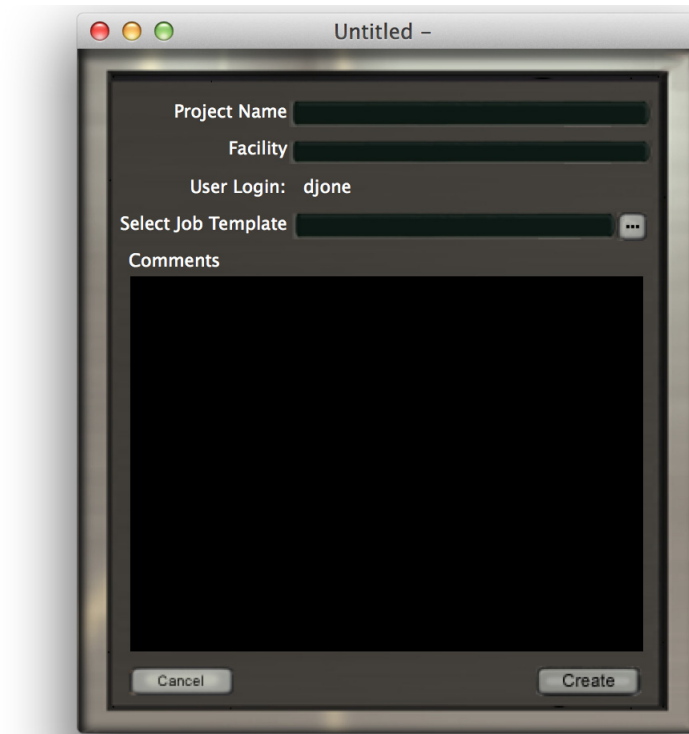
3. Enter information in the **Project Properties** window. See [Section 7.4.1](#).  
New projects require a project name. All other fields are optional.
4. Click **Create**.
5. When prompted in the **Save** dialog, do the following:
  - Save the project .dpp file with any name you choose at any desired location. The file name does not need to match the project name.
  - Click **OK**.

Once the project has been saved, the encoder opens the job list window and opens a job editor, allowing you to begin setting up the job for encoding.

Alternatively, you can create new jobs in an existing project (see [Section 7.7](#)), or open an existing job (see [Section 7.6.2](#)).

### 7.4.1 The Project Properties Window

In the **Project Properties** window, you can enter information for the new project.



**Figure 7-2** Project Properties Window

The encoder stores information in five fields in the **Project Properties** window.

- **Project Name:** This field is required to create a new project.
- **Facility** (optional): This field lets you designate the site where the encoding work was done. After creating a second project, the field will be automatically filled with the last value you entered.
- **User Login:** This field is automatically filled in with the current system's user name.
- **Select Job Template** (optional): If you browse to a .djt template file or enter its location here, the system uses that template for each new job you create in that project. Templates are always stored in a system template location. Clicking the browser button always takes you to that location automatically. Doing this makes templates more user-friendly for future use.
- **Comments** (optional): This field provides an area to enter information for later reference.

## 7.5 Changing Project Properties

After a project has been created and saved, you can view and change its properties in the **Project Properties** window.

To change project properties:

- Choose **File > Project Properties**.  
This command opens the job list window, in project view.

You can edit any of the four properties:

- **Project Name**
- **Facility**
- **Select Job Template**
- **Comments**

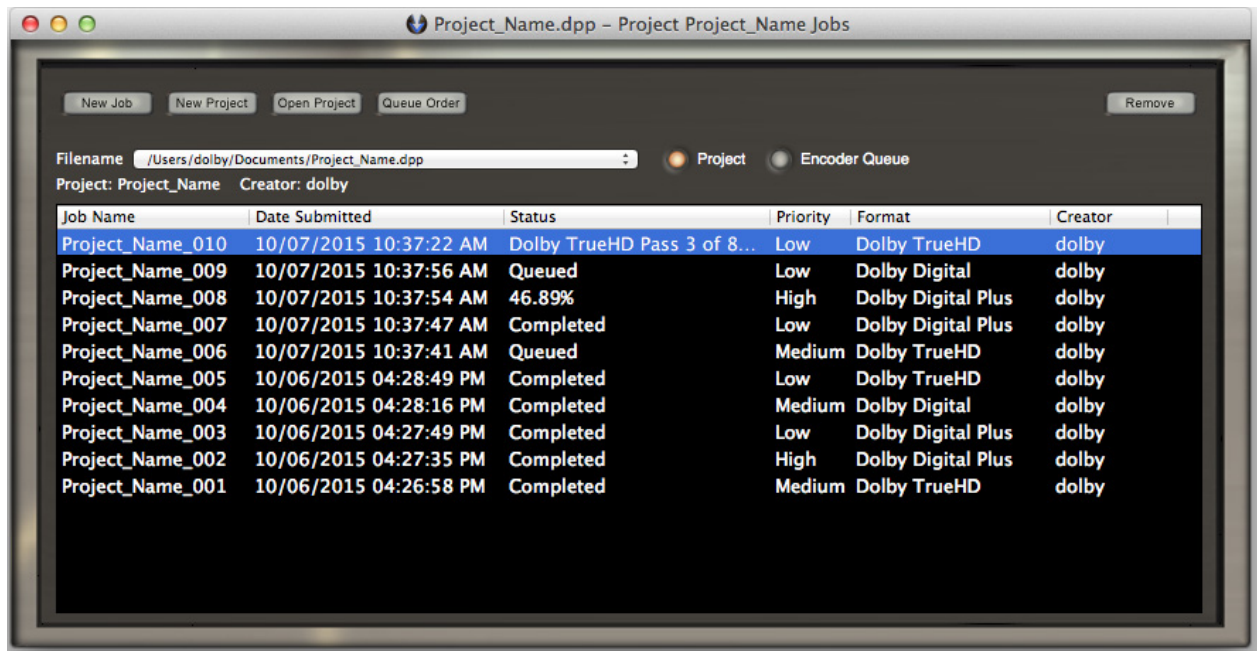


Figure 7-3 Job List: Project View

## 7.6 Viewing a Project and Its Jobs in Project View

The job list window includes a project view where you can view an existing project and all of its jobs. From this window, you can create new projects, new jobs to work on, or open existing projects.

To view an existing project, do one of the following:

- Choose **File > Open Project**, browse to an existing project, select it, and then click **Open** in the dialog.
- Choose **File > Open Recent** and select a recent project.
- Click the project icon on your computer.
- From the job list window, do either of the following:
  - Click the **Project** button, click the **Open Project** button, browse to an existing project, select it, and then click **Open** in the dialog.
  - Right-click on a project and click **Open**.
- From the Dolby Media Encoder window (the initial window that opens on startup), click the **Recent** drop-down menu and select a recent project, and then click the **Recent** button.
- If viewing the encoder queue in the job list window and you wish to view the jobs in the current project instead, click (enable) the **Project** button in the job list window.

### 7.6.1 Sorting Jobs in Project View

You can sort the order of jobs in project view. By default, the job list displays jobs in project view in alphabetical order by job name.

To change the order of jobs in project view:

- Clicking the heading of any column to sort by that value.
- Click **Queue Order** to return the page to the default sort order.

### 7.6.2 Opening or Removing Jobs in Project View

You can open or remove jobs in project view.

To open or remove a jobs:

1. In the job list, click **Project**.
2. Do one of the following:
  - Right-click on a job and select **Open** or **Remove**.
  - Select one of more jobs in the list, and then click the **Remove** button (located at the top of the job list window).

The encoder supports selecting one job by clicking (highlighting) it, or selecting multiple jobs by clicking (highlighting) a job and then selecting additional jobs by using the <Shift> or <Command> key.

## 7.7 Creating a New Job in an Existing Project

Once you have created a project, you can create additional jobs for it.

To create a new job in the current project, do one of the following:

- Choose **Job > New Job**.
- In the job list, click the **New Job** button.
- In the job editor window, click the **New Job** button.

---

---

# Encoding Configurations

The format and targets of the job determine the available encoding configurations. You set channel configurations in the **Channels** page.

## 8.1 Dolby TrueHD Encoding Configurations

When the format is Dolby® TrueHD, the system presents channel configuration and source-file options for Dolby Atmos® and eight-, six-, and two-channel presentations.



Dolby TrueHD Channels Page

Presentations define what each particular player will use to play back audio. An eight-channel player will use the eight-channel presentation, a six-channel player will use the six-channel presentation, and a two-channel player will use the two-channel presentation.

Each presentation can be one of the following:

- Independent (that is, independent source files can be used in that presentation)
- A copy of an independent presentation with fewer channels (if the bitstream does not include a Dolby Atmos presentation)
- A downmix of an independent presentation with more channels

The system defaults to an eight-channel independent presentation of **7.1 – L,R,C,LFE,Ls,Rs(Lrs,Rrs)** with the six-channel presentation being a downmix of the eight-channel presentation, and the two-channel presentation being a downmix of the six-channel presentation.



**Note:** Dolby Media Encoder v2.0 does not support encoding from PCM source audio for mono presentations to Dolby TrueHD bitstreams.

To encode from PCM source audio for mono presentations to Dolby TrueHD bitstreams, use the Encoder software installed with Suite v1.1.2 or earlier on a computer that does not have Dolby Media Encoder v2.0 or v1.8.x BETA installed on it.

## Eight-Channel Presentations for Dolby TrueHD

**Table 8-1** Dolby TrueHD Encoding Configurations (Eight-Channel Presentations)

| Encoding Configuration               | Encoding Scheme                                                                                         |
|--------------------------------------|---------------------------------------------------------------------------------------------------------|
| Copy of two channel                  | The eight-channel presentation will output the two-channel presentation.                                |
| Copy of six channel                  | The eight-channel presentation will output the six-channel presentation.                                |
| 6.1: L, R, C, LFE, Ls, Rs, Cs        | Left, Right, Center, LFE, Left Surround, Right Surround, Center Surround.                               |
| 7.1: L, R, C, LFE, Ls, Rs, Lrs, Rrs  | Left, Right, Center, LFE, Left Surround, Right Surround, Left Rear Surround, Right Rear Surround.       |
| 7.1: L, R, C, LFE, Ls, Rs, Lc, Rc    | Left, Right, Center, LFE, Left Surround, Right Surround, Left Center, Right Center.                     |
| 7.1: L, R, C, LFE, Ls, Rs, Lvh, Rvh  | Left, Right, Center, LFE, Left Surround, Right Surround, Left Vertical Height, Right Vertical Height.   |
| 7.1: L, R, C, LFE, Ls, Rs, Lw, Rw    | Left, Right, Center, LFE, Left Surround, Right Surround, Left Wide, Right Wide.                         |
| 7.1: L, R, C, LFE, Ls, Rs, Lts, Rts  | Left, Right, Center, LFE, Left Surround, Right Surround, Left Top Surround, Right Top Surround.         |
| Downmix from Dolby Atmos             | Use a 7.1 Lrs, Rrs downmix for the 8-channel presentation of a Dolby TrueHD bitstream with Dolby Atmos. |
| Dolby Atmos render to 7.1 (Lrs, Rrs) | Encode from a Dolby Atmos master file set to a 7.1 Dolby TrueHD bitstream without Dolby Atmos.          |



## Six-Channel Presentations for Dolby TrueHD

**Table 8-2** Dolby TrueHD Encoding Configurations (Six-Channel Presentations):

| Encoding Configuration    | Encoding Scheme                                                                       |
|---------------------------|---------------------------------------------------------------------------------------|
| Copy of two channel       | The six-channel presentation will output the two-channel presentation.                |
| Downmix of eight channel  | The six-channel presentation will output a downmix of the eight-channel presentation. |
| 5.1: L, R, C, LFE, Ls, Rs | Left, Right, Center, LFE, Left Surround, Right Surround.                              |
| 3.0: L, C, R              | Left, Center, Right.                                                                  |
| 3.1: L, C, R, LFE         | Left, Center, Right, LFE.                                                             |
| 4.0: L, R, Ls, Rs         | Left, Right, Left Surround, Right Surround.                                           |
| 5.0: L, R, C, Ls, Rs      | Left, Right, Center, Left Surround, Right Surround.                                   |

## Two-Channel Presentations for Dolby TrueHD

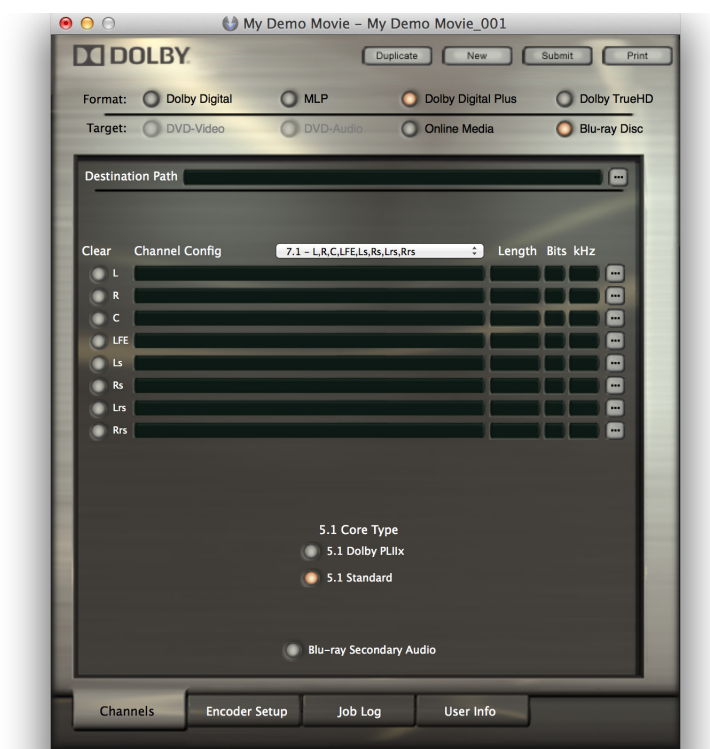
**Table 8-3** Dolby TrueHD Encoding Configurations (Two-Channel Presentations)

| Encoding Configuration   | Encoding Scheme                                                                       |
|--------------------------|---------------------------------------------------------------------------------------|
| Two channel              | Two-channel stereo, Left, Right (two-channel format defined in Dolby TrueHD encoder). |
| Downmix of six channel   | The two-channel presentation will output a downmix of the six-channel presentation.   |
| Downmix of eight channel | The two-channel presentation will output a downmix of the eight-channel presentation. |

## 8.2 Dolby Digital Plus Encoding Configurations

There are specific encoding configurations available when encoding material as Dolby Digital Plus.

For the Dolby Digital Plus™ format, the use of the Blu-ray Disc™ secondary audio option also determines which configurations will be available, as secondary audio must have 5.1 or fewer channels. If encoding secondary audio, select the option for it, and then select the encoding configuration.



**Figure 8-1** Dolby Digital Plus Channels Page

On the **Channels** page, you must choose a supported channel configuration, with outputs ranging from mono to 7.1, Dolby Atmos render to 7.1 (Lrs, Rrs) or 5.1 (Ls, Rs), and Dolby Atmos DD+ Bitstream (Dolby Atmos Dolby Digital Plus Bitstream), depending on the target. For example, with an online media target, you have a choice of two-channel configurations: stereo, Lt/Rt, or Dolby Headphone Encoded.



**Note:** The Dolby Media Encoder does not create an Lt/Rt or Dolby Headphone processed track. These channel configurations set metadata in the encoded stream so that a decoder can identify how the tracks should be played back.

With an online media target, you can also choose 5.1-channel Dolby Digital Plus or 5.1-channel Surround EX. When encoding traditional 5.1 Dolby Digital Plus programs, select the 5.1 channel Dolby Digital Plus configuration option. If you have source files with additional matrix-encoded channels already mixed into the Ls and Rs inputs, choose the **5.1 EX - L,R,C,LFE,Ls,Rs** option to set the proper metadata flag in the bitstream indicating content with additional matrix-encoded channels in the Ls and Rs channels, such as Dolby Pro Logic IIx or Dolby Surround EX content.

If you need to produce Dolby Digital Plus 5.1-channel files from 7.1 source files, the Dolby Media Encoder can do so automatically. It downmixes the source files first, and then encodes. Choose **5.1 - L,R,C,LFE,Ls,Rs (Lrs,Rrs)** for 7.1 source files. The channels in parentheses are the ones that will be matrix-encoded; these channels will be downmixed into the main 5.1 channel bed prior to encoding.

Choose the **Dolby Atmos DD+ Bitstream** option to encode from a Dolby Atmos master file set to a Dolby Digital Plus bitstream with Dolby Atmos. Choose the **Dolby Atmos render to 7.1 (Lrs, Rrs)** or **Dolby Atmos render to 5.1 (Ls, Rs)** options to encode from a Dolby Atmos master file set to a 7.1 or 5.1 Dolby Digital Plus bitstream without Dolby Atmos.

Additionally, these options enable **7.1 to 5.1 Downmix** options, which let you select the type of 7-to-5 downmix for the rear channels of the 7.1 or 5.1 core. For a description of 7.1 to 5.1 downmix options, see [Section 8.2.1](#).

**Table 8-4** Dolby Digital Plus Encoding Configurations

| Encoding Configuration                                               | Encoding Scheme                                                                                                                                                                                                                                                                     |
|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.0: Mono                                                            | Single Center channel. Available for online media or Blu-ray Disc secondary audio only.                                                                                                                                                                                             |
| 2.0: Stereo                                                          | Two-channel stereo Left, Right. Available for online media or Blu-ray Disc secondary audio only.                                                                                                                                                                                    |
| 2.0: Lt/Rt                                                           | Two-channel Left, Right (flagged as a matrix-encoded Lt/Rt signal). Available for online media only.                                                                                                                                                                                |
| 5.1: L, R, C, LFE, Ls, Rs                                            | Left, Right, Center, LFE, Left Surround, Right Surround. Available for online media or Blu-ray Disc secondary audio only.                                                                                                                                                           |
| 5.1 EX (Dolby Digital Surround EX): L, R, C, LFE, Ls, Rs             | Left, Right, Center, LFE, Left Surround, Right Surround (flagged as having a matrix-encoded rear channel). Available for online media only.                                                                                                                                         |
| 5.1: L, R, C, LFE, Ls, Rs (Lrs, Rrs)                                 | 5.1: Lrs, Rrs source files will be downmixed into the Ls/Rs prior to encoding. Available for online media only.<br><br>The encoder includes an option of standard (Lo, Ro) or Dolby Pro Logic® IIx downmixing.                                                                      |
| 2.0: Dolby Headphone encoded                                         | Two-channel flagged as a Dolby Headphone™ preprocessed mix. Available for online media only.<br><br>(Dolby Media Encoder does not create a Dolby Headphone preprocessed mix.)                                                                                                       |
| 3.0: L, C, R                                                         | Left, Center, Right. Available for online media or Blu-ray Disc secondary audio only.                                                                                                                                                                                               |
| 4.0: L, C, R, S                                                      | Left, Center, Right, mono Surround. Available for online media or Blu-ray Disc secondary audio only.                                                                                                                                                                                |
| 4.0: L, R, Ls, Rs                                                    | Left, Right, Left Surround, Right Surround. Available for online media or Blu-ray Disc secondary audio only.                                                                                                                                                                        |
| 4.1: L, R, Ls, Rs, LFE                                               | Left, Right, Left Surround, Right Surround, LFE. Available for online media or Blu-ray Disc secondary audio only.                                                                                                                                                                   |
| 4.1: L, C, R, S, LFE                                                 | Left, Center, Right, mono Surround, LFE. Available for online media or Blu-ray Disc secondary audio only.                                                                                                                                                                           |
| 5.0: L, R, C, Ls, Rs                                                 | Left, Right, Center, Left Surround, Right Surround. Available for online media or Blu-ray Disc secondary audio only.                                                                                                                                                                |
| 7.1: L, R, C, LFE, Ls, Rs, Lrs, Rrs                                  | Left, Right, Center, LFE, Left Surround, Right Surround, Left Rear Surround, Right Rear Surround.                                                                                                                                                                                   |
| 7.1: L, R, C, LFE, Ls, Rs, Lc, Rc                                    | Left, Right, Center, LFE, Left Surround, Right Surround, Left Center, Right Center.                                                                                                                                                                                                 |
| 7.1: L, R, C, LFE, Ls, Rs, Lvh, Rvh                                  | Left, Right, Center, LFE, Left Surround, Right Surround, Left Vertical Height, Right Vertical Height.                                                                                                                                                                               |
| 7.1: L, R, C, LFE, Ls, Rs, Lw, Rw                                    | Left, Right, Center, LFE, Left Surround, Right Surround, Left Wide, Right Wide.                                                                                                                                                                                                     |
| Dolby Atmos DD+ Bitstream (Dolby Atmos Dolby Digital Plus Bitstream) | Encode from a Dolby Atmos master file set to a Dolby Digital Plus bitstream with Dolby Atmos. This option enables 7.1 to 5.1 Downmix options, which let you select the type of 7-to-5 downmix for the rear channels of the 5.1 core.                                                |
| Dolby Atmos render to 5.1 (Ls, Rs)                                   | Encode from a Dolby Atmos master file set to a 5.1 Dolby Digital Plus bitstream without Dolby Atmos. Available for online media only.<br><br>This option enables 7.1 to 5.1 downmix options, which let you select the type of 7-to-5 downmix for the rear channels of the 5.1 core. |
| Dolby Atmos render to 7.1 (Lrs, Rrs)                                 | Encode from a Dolby Atmos master file set to a 7.1 Dolby Digital Plus bitstream without Dolby Atmos.<br><br>This option enables 7.1 to 5.1 downmix options, which let you select the type of 7-to-5 downmix for the rear channels of the 5.1 core.                                  |

### 8.2.1 Dolby Digital Plus Downmix Options on the Channel Page

The downmix options on the **Channels** page are different from those on the **Downmix** subpage of the **Encoder Setup** page. The downmix settings on the **Channels** page indicate source channels that will be downmixed before the 5.1 bitstream is encoded.

When you choose a channel configuration of **7.1 – L,R,C,LFE,Ls,Rs(Lrs,Rrs)**, **Dolby Atmos render to 7.1 (Lrs,Rrs)**, **Dolby Atmos render to 5.1 (Lrs,Rrs)**, or **Dolby Atmos DD+ Bitstream** with an Audio Core channel configuration of **7.1 – L,R,C,LFE,Lrs,Rrs** or **5.1 – L,R,C,LFE,Ls,Rs**, the encoder automatically creates the core 5.1 Dolby Digital bitstream channels for playback equipment that only supports 5.1 channels of output. The 5.1 core can be created with rear channels downmixed using the Standard (Lo, Ro) downmix or the Dolby Pro Logic IIx downmix by clicking the button for your preferred option. The default is Dolby Pro Logic IIx. For information on downmixing options and equations, see [Section 11.2](#).

## 8.3 Dolby Digital Encoding Configurations

There are specific encoding configurations available when encoding material as Dolby Digital.

On the **Channels** page, you must choose a supported channel configuration, with outputs ranging from mono to 5.1 to Dolby Atmos render to 5.1 (Ls, Rs). You have a choice of two-channel configurations: stereo, Lt/Rt, or Dolby Headphone Encoded.



**Note:** The Dolby Media Encoder does not create an Lt/Rt or Dolby Headphone processed track. These channel configurations set metadata in the encoded stream so that a decoder can identify how the tracks should be played back.

You can also choose 5.1-channel Dolby Digital or 5.1-channel Dolby Digital Surround EX. When encoding traditional 5.1 Dolby Digital programs, select the 5.1 channel Dolby Digital configuration option. If you have source files with additional matrix-encoded channels already mixed into the Ls and Rs inputs, choose the **5.1 EX – L,R,C,LFE,Ls,Rs** option to set the proper metadata flag in the bitstream indicating Dolby Digital Surround EX content.

If you need to produce Dolby Digital 5.1-channel files from 7.1 source files, the Dolby Media Encoder can do so automatically. It downmixes the source files first, and then encodes. Choose **5.1 – L,R,C,LFE,Ls,Rs (Lrs,Rrs)** for 7.1 source files. The channels in parentheses are the ones that will be matrix-encoded; these channels will be downmixed into the main 5.1 channel bed prior to encoding.

Choose the **Dolby Atmos render to 5.1 (Ls, Rs)** option to encode from a Dolby Atmos master file set to a 5.1 Dolby Digital bitstream without Dolby Atmos. This option lets you create a 5.1-channel Dolby Digital companion track for Blu-ray Discs carrying Dolby TrueHD with Dolby Atmos. The companion stream will match the six-channel presentation heard when playing back a Dolby TrueHD bitstream encoded from the same .atmos master source file. Additionally, this option enables **7.1 to 5.1 Downmix** options, which let you select the type of 7-to-5 downmix for the rear channels of the 5.1 core.

**Table 8-5** Dolby Digital Encoding Configurations

| Encoding Configuration               | Encoding Scheme                                                                                                                                                                                                                               |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.0: Mono                            | Single Center channel.                                                                                                                                                                                                                        |
| 2.0: Stereo                          | Two-channel stereo Left, Right.                                                                                                                                                                                                               |
| 2.0: Lt/Rt                           | Two-channel Left, Right (flagged as a matrix-encoded Lt/Rt signal).                                                                                                                                                                           |
| 5.1: L, R, C, LFE, Ls, Rs            | Left, Right, Center, Low-Frequency Effects, Left Surround, Right Surround.                                                                                                                                                                    |
| 5.1 Ex: L, R, C, LFE, Ls, Rs         | Dolby Digital Surround EX™: Left, Right, Center, LFE, Left Surround, Right Surround (flagged as having a matrix-encoded rear channel).                                                                                                        |
| 5.1: L, R, C, LFE, Ls, Rs (Lrs, Rrs) | 5.1: Lrs, Rrs source files will be downmixed into the Ls/Rs prior to encoding. An option of standard (Lo, Ro) or Dolby Pro Logic IIx downmixing is available.                                                                                 |
| 2.0: Dolby Headphone encoded         | Two-channel flagged as a Dolby Headphone preprocessed mix. (Dolby Media Encoder does not create a Dolby Headphone preprocessed mix.)                                                                                                          |
| 3.0: L, C, R                         | Left, Center, Right.                                                                                                                                                                                                                          |
| 4.0: L, C, R, S                      | Left, Center, Right, mono Surround.                                                                                                                                                                                                           |
| 4.0: L, R, Ls, Rs                    | Left, Right, Left Surround, Right Surround.                                                                                                                                                                                                   |
| 4.1: L, R, Ls, Rs, LFE               | Left, Right, Left Surround, Right Surround, LFE.                                                                                                                                                                                              |
| 4.1: L, C, R, S, LFE                 | Left, Center, Right, Mono Surround, LFE.                                                                                                                                                                                                      |
| 5.0: L, R, C, Ls, Rs                 | Left, Right, Center, Left Surround, Right Surround.                                                                                                                                                                                           |
| Dolby Atmos render to 5.1 (Ls, Rs)   | Encode from a Dolby Atmos master file set to a 5.1 Dolby Digital bitstream without Dolby Atmos.<br><br>This option enables 7.1 to 5.1 downmix options, which let you select the type of 7-to-5 downmix for the rear channels of the 5.1 core. |

### 8.3.1 Dolby Digital Downmix Options on the Channel Page

The downmix options on the **Channels** page are different from those on the **Downmix** subpage of the **Encoder Setup** page. The downmix settings on the **Channels** page indicate source channels that will be downmixed before the 5.1 bitstream is encoded.

When you choose a channel configuration of **5.1 – L,R,C,LFE,Ls,Rs(Lrs,Rrs)** or **Dolby Atmos render to 5.1 (Lrs,Rrs)**, you can choose either a Dolby Pro Logic IIx downmix, or a standard (Lo, Ro) 5.1 Dolby Digital downmix. **5.1 Dolby PLIIx** is the default. For information on downmixing options and equations, see [Section 11.2](#).

## 8.4 MLP Lossless Encoding Configurations

There are specific encoding configurations available when encoding material as MLP Lossless.

On the **Channels** page, you must choose a supported channel configuration, with outputs ranging from mono to 5.1.

**Table 8-6** MLP Lossless™ Encoding Configurations

| Encoding Configuration    | Encoding Scheme                                                            |
|---------------------------|----------------------------------------------------------------------------|
| 1.0: Mono                 | Single Center channel.                                                     |
| 2.0: Stereo               | Two-channel stereo Left, Right.                                            |
| 5.1: L, R, C, LFE, Ls, Rs | Left, Right, Center, Low-Frequency Effects, Left Surround, Right Surround. |
| 3.0: L, C, R              | Left, Center, Right.                                                       |
| 3.1: L, C, R, LFE         | Left, Center, Right, Low-Frequency Effects.                                |
| 4.0: L, R, Ls, Rs         | Left, Right, Left Surround, Right Surround.                                |
| 5.0: L, R, C, Ls, Rs      | Left, Right, Center, Left Surround, Right Surround.                        |

# Encoding Options

Each Dolby audio format has specialized encoding options.

- Dolby® TrueHD. See [Section 9.1](#).
- Dolby Digital Plus™. See [Section 9.2](#).
- Dolby Digital. See [Section 9.3](#).
- MLP Lossless™. See [Section 9.4](#).

## 9.1 Dolby TrueHD Encoding Options

There are specific encoding options available when encoding material as Dolby TrueHD for online media or Blu-ray Disc™.

You access these options in the **Encoder Setup** subpages:

- [Encoder Settings](#)
- [Downmix](#)
- [Preprocessing](#)
- [Timecode](#)

### 9.1.1 Encoder Settings

Most Dolby TrueHD encoder setup is done separately for each presentation. Any presentation that is a copy of another has no other setup features.



---

**Note:** Having more than one independent presentation increases the final file size and data rate.

---

## Dolby Atmos Presentation

**Table 9-1** Dolby Atmos Presentation Encoder Settings

| Parameter                         | Range                                                                                                                                                         |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dialogue level                    | The default is <b>-31</b> . The valid range is <b>-1</b> to <b>-31</b> .                                                                                      |
| Late-night listening mode profile | The default is <b>Film Light</b> . There are four other options. For details, see <a href="#">Dynamic Range (Late-Night Listening Mode Profile) Presets</a> . |

## Eight-Channel Presentation

**Table 9-2** Eight-Channel Presentation Encoder Settings

| Parameter                         | Range                                                                                                                                                         |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dialogue level                    | The default is <b>-31</b> . The valid range is <b>-1</b> to <b>-31</b> .                                                                                      |
| Late-night listening mode profile | The default is <b>Film Light</b> . There are four other options. For details, see <a href="#">Dynamic Range (Late-Night Listening Mode Profile) Presets</a> . |

## Six-Channel Presentation

**Table 9-3** Six-Channel Presentation Encoder Settings

| Parameter                         | Range                                                                                                                                                                                                                                                                                                       |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dialogue level                    | The default is <b>-31</b> . The valid range is <b>-1</b> to <b>-31</b> .                                                                                                                                                                                                                                    |
| Dolby Digital Surround EX         | The default is <b>Dolby Digital EX Encoded</b> , since Dolby PLIIx is enabled by default and this flag is used to indicate that both Surround EX and Pro Logic IIx matrix-encoded content is present in the rear channels. Other options are <b>Not Indicated</b> and <b>Not Dolby Digital EX Encoded</b> . |
| Late-night listening mode profile | The default is <b>Film Light</b> . There are four other options. For details, see <a href="#">Dynamic Range (Late-Night Listening Mode Profile) Presets</a> .                                                                                                                                               |

## Two-Channel Presentation

**Table 9-4** Two-Channel Presentation Encoder Settings

| Parameter                         | Range                                                                                                                                                           |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dialogue level                    | The default is <b>-31</b> . The valid range is <b>-1</b> to <b>-31</b> .                                                                                        |
| Two-channel format                | The default is <b>Dolby Surround Encoded</b> . Other options are <b>Dual</b> , <b>Stereo</b> (not Dolby Surround encoded), and <b>Dolby Headphone Encoded</b> . |
| Late-night listening mode profile | The default is <b>Film Light</b> . There are four other options. For details, see <a href="#">Dynamic Range (Late-Night Listening Mode Profile) Presets</a> .   |
| Late-night listening default      | <b>On</b> or <b>Off</b> . (Available when presentation is a downmix of 6- or 8-channel)                                                                         |

## Run Verification After Encode Button

For each Dolby TrueHD encoding job, Dolby Media Encoder can run a verification pass.

Generally, you will want to include verification by enabling the Run Verification After Encode button. This button is enabled by default.

Deselect this button to skip the verification of the Dolby TrueHD encoding process.



### 9.1.2 Downmix

The downmix options are configurable. For the eight- to six-channel downmix, you can choose:

- Dolby Pro Logic® IIx
- Standard
- Custom

For the six- to two-channel downmix, you can choose:

- Standard
- Custom



**Note:** The downmix equations are the same as the ones used for the options with the same descriptions in the Dolby Digital and Dolby Digital Plus **Channels** pages. For information on downmixing options and equations, see [Section 11.2](#).

If you choose custom, you must click **Open** at the bottom of the subpage to open the **Custom Downmix** page. On this page, you can choose the channels you wish to include, as well as their levels and phase settings.

Each channel includes a **Phase** button to invert the phase of the signal.

Click **Close** at the bottom of the subpage to close the **Custom Downmix** page

### 9.1.3 Preprocessing

The **Preprocessing** subpage includes options for each independent presentation, as well as for encoding from Dolby Atmos master file sets.

**Table 9-5** Dolby TrueHD Preprocessing Options for Each Independent Presentation

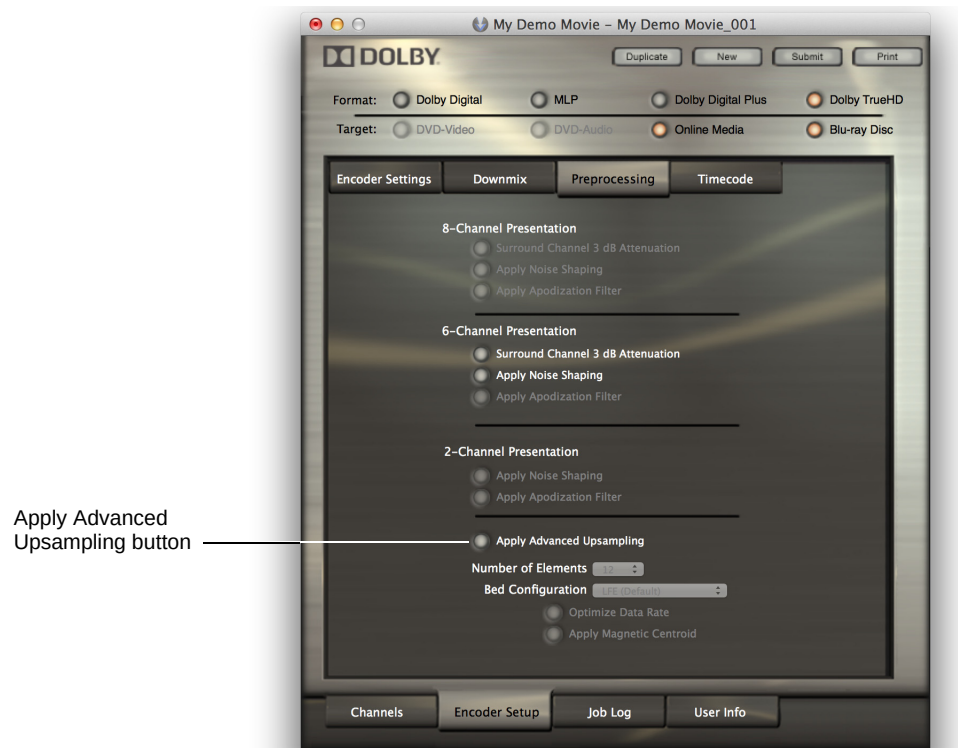
| Parameter                         | Range                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Surround channel 3 dB attenuation | <b>On</b> or <b>Off</b> . For details, see <a href="#">Surround Channel 3 dB Attenuation</a> .<br><b>Note:</b> This parameter does not apply to the two-channel presentation.                                                                                                                                                                                                                                 |
| Apply apodization filter          | <b>On</b> or <b>Off</b> .<br>When applied, the apodization filter assists in minimizing the ringing effect that can occur in some analog-to-digital and digital-to-analog filters. Apodization filters can aid in decoding efficiency and improve the response of the audio through many A/D converters. The apodization filter is applicable only to source files with a sampling rate of at least 88.2 kHz. |
| Allow noise shaping               | <b>On</b> or <b>Off</b> .<br>When applied, noise shaping reduces the entropy of the low-level noise in the least-significant bits of a 24-bit source file. Minimizing this noise can aid in coding efficiency without affecting audio quality. Noise shaping will have no effect on 16-bit source files.                                                                                                      |

**Table 9-6** Dolby TrueHD Preprocessing Options for Dolby Atmos Master File Sets

| Parameter               | Range                                                                                                                                                                                                                                                                                                             |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Number of elements      | This option sets the number of clustered signals plus bed signals of Dolby Atmos® material to be output by spatial coding when encoding rendered 7.1 audio from a Dolby master file set to channel-based Dolby Digital Plus bitstreams.                                                                           |
| Bed configuration       | Choices include: LFE (Default); LFE,C; LFE,L,C,R; LFE,L,C,R,Ls,Rs; LFE,Ls,Rs,Lrs,Rrs; LFE,Ls,Rs<br><br>This option sets the bed configuration for spatial coding. By default, this option is set to LFE (default). Leave this option at its default setting, unless Dolby support recommends a different setting. |
| Optimize data rate      | Enabled or disabled. Default is enabled.<br><br>When enabled, intermediate signals in the signal path are treated with additional quantization when encoding from a Dolby Atmos master file set to Dolby TrueHD. This can reduce the peak and average data rates by 10-15% for typical content.                   |
| Apply magnetic centroid | When enabled, helps ensure consistent behavior for demanding content when encoding from a Dolby master file set. By default, this option is enabled (lit). Leave this option at its default setting, unless Dolby support recommends a different setting.                                                         |

### Apply Advanced Upsampling Button (.wav Source Files Only)

The **Apply Advanced Upsampling** button engages an upsampler that is applied to all .wav files loaded into the presentation fields. The option is available for 44.1 or 48 kHz audio sample rates, when encoding from Dolby TrueHD without Dolby Atmos. When selected, 44.1 kHz audio is upsampled to 88.2 kHz, and 48 kHz audio is upsampled to 96 kHz.



**Figure 9-1** Advanced Upsampling Option



**Note:** The Blu-ray Disc specification does not support 44.1, 88.2, or 176.4 kHz.

A specialized apodizing filter is tied to this high-quality upsampler. It is designed to mask any pre-ringing due to brick-wall anti-aliasing filters in the upsampled audio. When the pre-ringing artifact is masked, the original audio feels more open, and has a clearer high-frequency depth and deeper high-frequency decay.

#### 9.1.4 Timecode

The **Timecode** subpage provides the same display and options for Dolby Digital, Dolby Digital Plus, and Dolby TrueHD.

For more information, see [Section 9.6](#).

## 9.2 Dolby Digital Plus Encoding Options

There are specific features and options available when encoding material as Dolby Digital Plus for online media or Blu-ray Disc.

You access these options in the **Encoder Setup** subpages:

- [Encoder Settings](#)
- [Downmix](#)
- [Preprocessing](#)
- [Timecode](#)

### 9.2.1 Encoder Settings

When the format is Dolby Digital Plus, the **Encoder Settings** subpage provides settings for dialogue level (also known as dialnorm), data rate, and other options.

**Table 9-7** Dolby Digital Plus Encoder Settings

| Parameter                        | Range                                                                                                                                                                                                                                                                                        |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dialogue level                   | The default is <b>-31</b> . The range is <b>-1</b> to <b>-31</b> .                                                                                                                                                                                                                           |
| Data rate                        | Based on the selected format, target, and number of input channels, the system automatically selects the most common data rate and generates a list of alternatives. To change the system default, choose a rate from the <b>Data Rate</b> drop-down menu.                                   |
| Bitstream mode                   | The default is <b>Complete Main</b> . There are seven other options (for online media targets with up to 5.1 configurations): <b>Music &amp; Effects</b> , <b>Visually Impaired</b> , <b>Hearing Impaired</b> , <b>Dialogue</b> , <b>Commentary</b> , <b>Emergency</b> , <b>Voice Over</b> . |
| Line mode                        | The default is <b>Film Light</b> . There are five options.<br>For details, see <a href="#">Dynamic Range (Late-Night Listening Mode Profile) Presets</a> .                                                                                                                                   |
| RF mode                          | The default is <b>Film Light</b> . There are five options.<br>For details, see <a href="#">Dynamic Range (Late-Night Listening Mode Profile) Presets</a> .                                                                                                                                   |
| Blu-ray secondary audio settings | There are three parameters. For details, see <a href="#">Blu-ray Disc Secondary Audio</a> .                                                                                                                                                                                                  |
| Optional user inserted ID number | Default is <b>0</b> . Up to five digits can be entered.                                                                                                                                                                                                                                      |

When encoding from a Dolby Atmos master file set to create a Dolby Digital Plus bitstream for online media, there are additional data rate options available for 7.1 multichannel configurations that include Lrs and Rrs: 384, 448, 504, 576, 640, and 704 kbps.

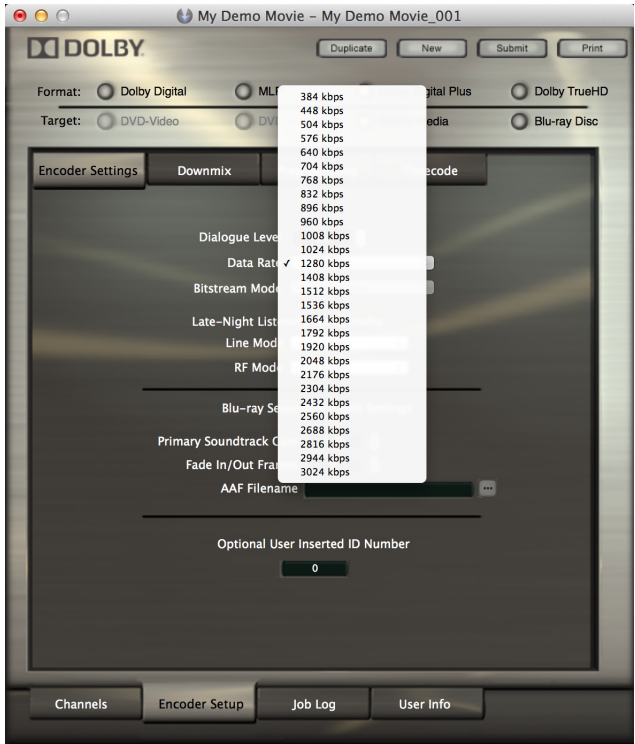


Figure 9-2 Data Rate Options in the Encoder Settings Subpage



**Note:** Data rates below 640 kbps are not supported when encoding from an .atmos master file set to Dolby Digital Plus with a 7.1 core channel configuration that includes Lv<sub>h</sub> and Rv<sub>h</sub> (left and right “vertical height”).

Options are based on the channel configuration of the backward-compatible presentation (as set with the **Audio Core Channel Configuration** drop-down menu).

Blu-ray Disc Secondary Audio

One of the most compelling features of the Blu-ray Disc format is its ability to include secondary audio tracks for interactive features. This feature set, called BonusView (or Blu-ray Profile 1.1), supports up to 5.1 channels of secondary audio that lets viewers enjoy, for instance, a director’s commentary accompanying a movie, or a picture-in-picture feature with its own surround soundtrack.

BonusView support is required in all new Blu-ray Disc players. Dolby Media Encoder supports BonusView with three additional options.

Table 9-8 Blu-ray Disc Secondary Audio Options

| Parameter               | Range                                              |
|-------------------------|----------------------------------------------------|
| Primary soundtrack gain | –inf (infinity) or –50 to +12 dB. Default is 0 dB. |
| Fade in/out frames      | 0 to 10. Default is 0.                             |
| AAF file name           | A valid name for the .aaf secondary audio file.    |

The format for Dolby Digital Plus Blu-ray Disc secondary audio is .ec3.

To access **Blu-ray Secondary Audio** options, you must first select the **Blu-ray Secondary Audio** option on the **Channels** page.

You can import .aaf automation metadata from Pro Tools® to embed dynamic mix metadata. Importing an .aaf file negates the primary soundtrack gain feature. See [Using an Advanced Authoring Format Automation File](#).

## Data Rates

The secondary audio encoder supports the following data rates specified by Blu-ray Disc:

- 224–256 kbps for 5.1 audio
- 96–256 kbps for 2.0 audio
- 56–256 kbps for 1.0 audio

## Metadata Processes

You can embed metadata for the secondary audio using either of two processes:

- Choosing a fixed primary soundtrack gain and a number of fade-in/fade-out frames
- Importing metadata from an .aaf file

## Using Primary Soundtrack Gain

The primary soundtrack gain value tells the player how much attenuation to apply to the primary soundtrack when secondary audio is used. This is a static value that will not change throughout the encode. The range is **–inf dB** or **–50 dB to +12 dB** (in 1 dB increments).

The fade-in/fade-out frames value sets the number of frames over which secondary audio fades in and out. The range is **0** to **10**.

## Using an Advanced Authoring Format Automation File

If you include an .aaf file, the system deactivates primary soundtrack gain and fade-in/fade-out frames and instead inserts dynamic gain metadata from the selected .aaf automation file.

There are important conditions that .aaf files must meet to be usable.

The .aaf file may contain up to 5.1 channels depending on the size of the presentation.

If you use a single-channel .aaf file, that single automation channel is applied to all the secondary audio channels. It can apply up to 50 dB of attenuation.

If you use a multichannel .aaf file:

- The .aaf file must contain a separate automation track for each audio channel.
  - The number of channels in the .aaf file must match the number of channels of audio in the secondary stream.
  - The channel order in the .aaf file must match the order shown in the following table.

**Table 9-9** .aaf File Channel Order

| Channel | File |
|---------|------|
| 1       | L    |
| 2       | R    |
| 3       | C    |
| 4       | LFE  |
| 5       | Ls   |
| 6       | Rs   |

- The file may apply up to 28 dB of attenuation per channel.
- A secondary audio track cannot be mixed in a Blu-ray Disc player if the number of audio channels in a secondary audio track exceeds the number of audio channels in a primary audio track.
- Secondary and primary audio start times and frame rate must match.
- Exported .aaf file start time and frame rate must match the secondary audio files in which they will be embedded.

The secondary audio encoder scans your .aaf file prior to encoding and advises you if you have exceeded any of the listed conditions. You can then either remix your .aaf file to comply or agree to have the Dolby Media Encoder limit your automation to the Blu-ray Disc specification attenuation restrictions.



**Note:** Attenuation levels, channel configurations, and fade resolution limitations are determined by the Blu-ray Disc specification.

If your .aaf file is longer than the secondary audio source files, Dolby Media Encoder will trim off the excess data.

If your .aaf file is shorter than the secondary audio source files, Dolby Media Encoder will use the last attenuation value in the file through the end of the audio encode.

For detailed information on preparing secondary audio and automation in Pro Tools, see [Preparing Secondary Audio and Automation in Pro Tools](#).

## 9.2.2 Downmix

The **Downmix** subpage defines metadata to be included in the encoded stream for later use by the decoder during playback. This metadata instructs a decoder how to downmix for a two-channel output.

The options defined on the **Downmix** subpage are different from the downmix options on the **Channels** page.

The available options are listed in [Table 9-10](#).

**Table 9-10** Dolby Digital Plus Bitstream Information Options

| Parameter                | Range                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lt/Rt center             | The default mix level for the center signal into L and R channels of a Lt/Rt downmix is <b>-3 dB</b> . There are seven choices between <b>-6</b> and <b>+3 dB</b> .                                                                                                                                                                                   |
| Lt/Rt surround           | The default mix level for the surround signals into L and R channels of an Lt/Rt downmix is <b>-3 dB</b> . There are five choices between <b>-infinity</b> and <b>-1.5 dB</b> .                                                                                                                                                                       |
| Lo/Ro center             | The default mix level for the center signal into L and R channels of a Lo/Ro downmix is <b>-3 dB</b> . There are seven choices between <b>-6</b> and <b>+3 dB</b> .                                                                                                                                                                                   |
| Lo/Ro surround           | The default mix level for the surround signals into L and R channels of a Lo/Ro downmix is <b>-3 dB</b> . There are five choices between <b>-infinity</b> and <b>-1.5 dB</b> .                                                                                                                                                                        |
| Preferred stereo downmix | <b>Pro Logic II</b> , the default for online media, instructs the decoder to perform a Lt/Rt downmix compatible with Dolby <b>Pro Logic II</b> , <b>IIx</b> , and <b>IIz</b> upmixing. <b>Surround</b> , the default for Blu-ray, Disc, is for conventional Pro Logic® Lt/Rt downmixing. Other options are <b>Not Indicated</b> , and <b>Stereo</b> . |

### 9.2.3 Preprocessing

The **Preprocessing** subpage determines whether several optional processes are applied to your source files as part of the encoding process.

**Table 9-11** Dolby Digital Plus Preprocessing Options

| Parameter                              | Range                                                                                                                                                                                                                                          |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LFE lowpass filter                     | <b>On</b> or <b>Off</b> . For details, see <a href="#">LFE Lowpass Filter</a> .                                                                                                                                                                |
| Surround channel 90-degree phase shift | <b>On</b> or <b>Off</b> . For details, see <a href="#">Surround Channel 90-Degree Phase-Shift</a> .                                                                                                                                            |
| Surround channel 3 dB attenuation      | <b>On</b> or <b>Off</b> . For details, see <a href="#">Surround Channel 3 dB Attenuation</a> .                                                                                                                                                 |
| Bed configuration                      | LFE (Default) is the default. There are six options.                                                                                                                                                                                           |
| Number of elements                     | This option sets the number of clustered signals plus bed signals of Dolby Atmos® material to be output by spatial coding when encoding rendered 7.1 or 5.1 audio from a Dolby master file set to channel-based Dolby Digital Plus bitstreams. |
| Apply magnetic centroid                | Enabled (lit) or disabled.                                                                                                                                                                                                                     |

#### LFE Lowpass Filter

The **LFE Lowpass Filter** option is available only when the LFE channel is enabled.

Click **LFE Lowpass Filter** to apply a 120 Hz lowpass filter to the LFE input channel. This filter should always be enabled unless you are absolutely sure that your LFE source has no audio above 120 Hz. It is best to leave the filter on, as there is no adverse effect from doing so.



The LFE lowpass filter is an eighth-order lowpass filter with unity gain at low frequencies and a steep rolloff at the upper edge of the LFE bandwidth (–3 dB above 120 Hz).

### Surround Channel 90-Degree Phase-Shift

The **Surround Channel 90-Degree Phase-Shift** option is available only when surround channels are present in the source.

Click **Surround Channel 90-Degree Phase-Shift** to ensure the best experience for users who decode and listen to your 5.1-channel program downmixed to a 2/0 Lt/Rt signal. The phase shift reduces undesirable signal cancellation in the stereo presentation. The phase shift also makes it easier for many upmixers to extract the rear-channel information, resulting in better separation between the front and rear channels. It is especially useful if there are similar sounds in the L, R, Ls and Rs source channels.

If you are satisfied with your surround-decoded downmix without the use of this filter—or if, for example, your program is a multichannel music program with discrete instrumentation hard-panned in all channels—it may be acceptable to turn this feature off.

### Surround Channel 3 dB Attenuation

The **Surround Channel 3 dB Attenuation** option is available only when surround channels are present in the source and the source is a set of PCM channel signals.

Click **Surround Channel 3 dB Attenuation** to apply a 3 dB attenuation to the surround channel source files prior to encoding.

This is appropriate when encoding a multichannel soundtrack created in a room with film-style calibration for consumer home theater playback. Channel-based cinema soundtrack surround channels have been traditionally recorded 3 dB louder relative to the front channels in order to account for cinema calibration standards, which require that the surround loudspeakers are set 3 dB lower than the front (screen) channels. Home theater surround speakers are calibrated differently, so that all speakers play back at the same level. This option is not available when encoding from a Dolby Atmos master file set, because Dolby Atmos content is mixed for both the cinema and the home with all speakers calibrated to the same SPL.

### Bed Configuration

The **Bed Configuration** option sets the bed configuration for spatial coding. By default, this option is set to LFE (Default). Leave this option at its default setting, unless Dolby support recommends a different setting.

Choices include:

- LFE (Default)
- LFE, C
- LFE, L, C, R
- LFE, L, C, R, Ls, Rs
- LFE, Ls, Rs, Lrs, Rrs
- LFE, Ls, Rs

## Apply Magnetic Centroid

The **Apply Magnetic Centroid** option, when enabled, helps ensure consistent behavior for demanding content when encoding from a Dolby master file set. By default, this option is enabled (lit). Leave this option at its default setting, unless Dolby support recommends a different setting.

### 9.2.4 Timecode

The **Timecode** subpage provides the same display and options for Dolby Digital, Dolby Digital Plus, and Dolby TrueHD.

For more information, see [Section 9.6](#).

## 9.3 Dolby Digital Encoding Options

There are specific features and options available when encoding material as Dolby Digital for a DVD or Blu-ray Disc.

You access these options in the **Encoder Setup** subpages:

- [Encoder Settings](#)
- [Downmix](#)
- [Preprocessing](#)
- [Timecode](#)

### 9.3.1 Encoder Settings

The **Encoder Settings** subpage provides settings for dialogue level (also known as dialnorm), data rate, and other options, as listed in [Table 9-12](#).

**Table 9-12** Dolby Digital Encoder Settings

| Parameter      | Range                                                                                                                                                                                                                                                    |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dialogue level | The default is <b>-31</b> . The range is <b>-1</b> to <b>-31</b> .                                                                                                                                                                                       |
| Data rate      | Based on the chosen format, target, and number of input channels, the system automatically selects the most common data rate and generates a list of alternatives. To change the system default, choose a rate from the <b>Data Rate</b> drop-down menu. |
| Bitstream mode | The default is <b>Complete Main</b> . There are seven other options: <b>Music &amp; Effects</b> , <b>Visually Impaired</b> , <b>Hearing Impaired</b> , <b>Dialogue</b> , <b>Commentary</b> , <b>Emergency</b> , <b>Voice Over</b> .                      |
| Line mode      | The default is <b>Film Light</b> . There are five options.<br>For details, see <a href="#">Dynamic Range (Late-Night Listening Mode Profile) Presets</a> .                                                                                               |
| RF mode        | The default is <b>Film Light</b> . There are six options.<br>For details, see <a href="#">Dynamic Range (Late-Night Listening Mode Profile) Presets</a> .                                                                                                |

### 9.3.2 Downmix Options

The **Downmix** subpage defines metadata to be included in the encoded stream for later use by the decoder during playback. This metadata instructs a decoder how to downmix a multichannel bitstream for two-channel output.

The options defined on the **Downmix** subpage are different from the downmix options on the **Channels** page. The Lx/Rx mix-level parameters let the content creator set the levels at which the Center and Surround signals will be mixed into the L and R outputs when downmixing is active in a decoder product. The levels can be set independently for the surround-encoded (Lt/Rt) and stereo (Lo/Ro) downmix types. The **Preferred Stereo Downmix** parameter instructs a decoder that is set to the **Auto** downmix selection to enable the producer's preferred choice when downmixing.

The mix-level metadata options are defined in the following table.

**Table 9-13** Dolby Digital Downmix Options

| Parameter                | Range                                                                                            |
|--------------------------|--------------------------------------------------------------------------------------------------|
| Lt/Rt center             | The default is <b>-3</b> dB. There are seven choices between <b>-6</b> and <b>+3</b> dB.         |
| Lt/Rt surround           | The default is <b>-3</b> dB. There are five choices between <b>-infinity</b> and <b>-1.5</b> dB. |
| Lo/Ro center             | The default is <b>-3</b> dB. There are seven choices between <b>-6</b> and <b>+3</b> dB.         |
| Lo/Ro surround           | The default is <b>-3</b> dB. There are five choices between <b>-infinity</b> and <b>-1.5</b> dB. |
| Preferred stereo downmix | The default is <b>Surround</b> . Other options are <b>Not Indicated, Stereo</b> .                |

### 9.3.3 Preprocessing

The **Preprocessing** subpage determines whether several optional processes are applied to your source files as part of the encoding process.

The following options are provided:

- [LFE Lowpass Filter](#)
- [Surround Channel 90-Degree Phase-Shift](#)
- [Surround Channel 3 dB Attenuation](#)

**Table 9-14** Dolby Digital Preprocessing Options

| Parameter                              | Range                                                |
|----------------------------------------|------------------------------------------------------|
| LFE lowpass filter                     | Enabled (lit) or disabled.                           |
| Surround channel 90-degree phase shift | Enabled (lit) or disabled.                           |
| Surround channel 3 dB attenuation      | Enabled (lit) or disabled.                           |
| Bed configuration                      | LFE (Default) is the default. There are six options. |
| Apply magnetic centroid                | Enabled (lit) or disabled.                           |

#### LFE Lowpass Filter

The **LFE Lowpass Filter** option is available only when the LFE channel is enabled.

Click **LFE Lowpass Filter** to apply a 120 Hz lowpass filter to the LFE input channel. This filter should always be enabled unless you are absolutely sure that your LFE source has no audio above 120 Hz. It is best to leave the filter on, as there is no adverse effect from doing so.

The LFE lowpass filter is an eighth-order lowpass filter with unity gain at low frequencies and a steep rolloff at the upper edge of the LFE bandwidth (–3 dB above 120 Hz).

### Surround Channel 90-Degree Phase-Shift

The **Surround Channel 90-Degree Phase-Shift** option is available only when surround channels are present in the source.

Click **Surround Channel 90-Degree Phase-Shift** to ensure the best experience for users who decode and listen to your 5.1-channel program downmixed to a 2/0 Lt/Rt signal. The phase shift reduces undesirable signal cancellation in the stereo presentation. The phase shift also makes it easier for many upmixers to extract the rear-channel information, resulting in better separation between the front and rear channels. It is especially useful if there are similar sounds in the L, R, Ls and Rs source channels.

If you are satisfied with your surround-decoded downmix without the use of this filter—or if, for example, your program is a multichannel music program with discrete instrumentation hard-panned in all channels—it may be acceptable to turn this feature off.

### Surround Channel 3 dB Attenuation

The **Surround Channel 3 dB Attenuation** option is available only when surround channels are present in the source and the source is a set of PCM channel signals.

Click **Surround Channel 3 dB Attenuation** to apply a 3 dB attenuation to the surround channel source files prior to encoding.

This is appropriate when encoding a multichannel soundtrack created in a room with film-style calibration for consumer home theater playback. Channel-based cinema soundtrack surround channels have been traditionally recorded 3 dB louder relative to the front channels in order to account for cinema calibration standards, which require that the surround loudspeakers are set 3 dB lower than the front (screen) channels. Home theater surround speakers are calibrated differently, so that all speakers play back at the same level. This option is not available when encoding from a Dolby Atmos master file set, because Dolby Atmos content is mixed for both the cinema and the home with all speakers calibrated to the same SPL.

### Bed Configuration

The **Bed Configuration** option sets the bed configuration for spatial coding. This parameter is only applicable when rendering 5.1 channels of audio from a Dolby Atmos master file for encoding to a 5.1 channel Dolby Digital bitstream. By default, this option is set to LFE (Default). Leave this option at its default setting, unless Dolby support recommends a different setting.

Choices include:

- LFE (Default)
- LFE, C
- LFE, L, C, R
- LFE, L, C, R, Ls, Rs
- LFE, Ls, Rs, Lrs, Rrs
- LFE, Ls, Rs

## Apply Magnetic Centroid

The **Apply Magnetic Centroid** option, when enabled, helps ensure consistent behavior for demanding content when encoding from a Dolby master file set. By default, this option is enabled (lit). Leave this option at its default setting, unless Dolby support recommends a different setting.

### 9.3.4 Timecode

Use the **Timecode** subpage to enable or disable timecode embedding, as well as to choose the timecode format and starting time for the embedded code. You can also specify a timecode range to encode when encoding just a portion of the input file, and you can choose to prepend silence to the beginning of the file.

The **Timecode** subpage provides the same display and options for Dolby Digital, Dolby Digital Plus, and Dolby TrueHD.

For more information, see [Section 9.6](#).

## 9.4 MLP Lossless Encoding Options

There are specific features and options available when encoding material as MLP Lossless for DVD-Audio.

You access these options in the **Encoder Setup** subpages:

- [Encoder Settings](#)
- [Downmix](#)
- [Preprocessing](#)
- [Timecode](#)

### 9.4.1 Encoder Settings

The MLP format has a single field in the **Encoder Settings** subpage: **Restart Point in Access Units**.

This setting lets you specify the number of access units between restart points. The default spacing for MLP Lossless audio is the maximum spacing, which uses one restart point every 32 access units, equivalent to a time interval of 27 ms. Selecting a shorter interval between restart points will allow decoding from midstream (after a pause, fast-forward or rewind operation) to start in less than 27 ms, but the data rate of the MLP Lossless stream will increase. An interval of 32 access units is appropriate for most applications.

### 9.4.2 Downmix

On the **Downmix** subpage, you can select the channels to be included in the two-channel downmix and specify a separate attenuation for each channel.

Each channel includes a **Phase** button to invert the phase of the signal.

### 9.4.3 Preprocessing

The **Preprocessing** subpage for encoding to the MLP format includes two options.

**Table 9-15** Dolby TrueHD Preprocessing Options

| Parameter                         | Range                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Surround channel 3 dB attenuation | <b>On</b> or <b>Off</b> . For details, see <a href="#">Surround Channel 3 dB Attenuation</a> .                                                                                                                                                                                                                                                                                                                |
| Apply apodization filter          | <b>On</b> or <b>Off</b> .<br>When applied, the apodization filter assists in minimizing the ringing effect that can occur in some analog-to-digital and digital-to-analog filters. Apodization filters can aid in decoding efficiency and improve the response of the audio through many A/D converters. The apodization filter is applicable only to source files with a sampling rate of at least 88.2 kHz. |

### 9.4.4 Timecode

On the **Timecode** sub page for encoding to the MLP format, you can choose to specify a time range to encode.

**Table 9-16** MLP Lossless Timecode Options

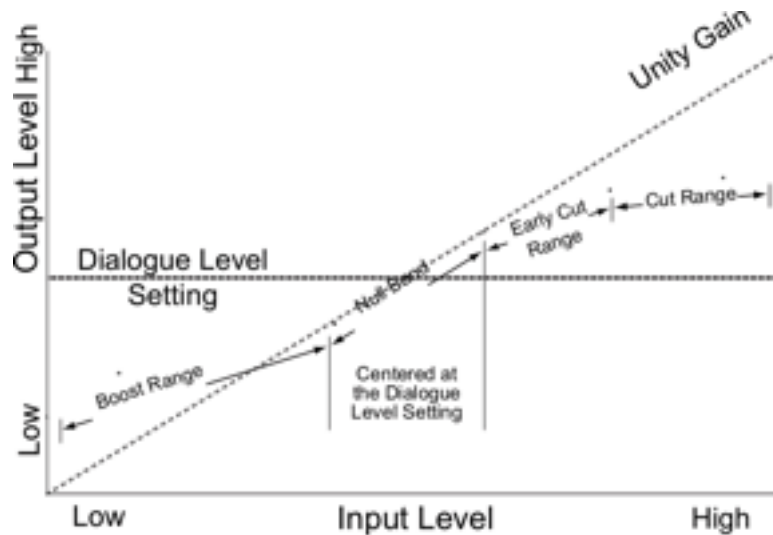
| Parameter                    | Range                                                                                   |
|------------------------------|-----------------------------------------------------------------------------------------|
| Specify time range to encode | <b>On</b> or <b>Off</b> .                                                               |
| Encode start                 | Any valid time value.                                                                   |
| Encode end                   | Any valid time value.                                                                   |
| Frame rate                   | The default is <b>23.976</b> . There are nine other options, including real time.       |
| Prepend silence              | Set a number of frames and milliseconds of silence to add at the beginning of the file. |

## 9.5 Dynamic Range (Late-Night Listening Mode Profile) Presets

Dynamic range presets let you choose the compression characteristic that could be applied to a Dolby TrueHD, Dolby Digital Plus, or Dolby Digital bitstream during decoding. These compression presets aid playback for two-channel downmix overload protection and late-night listening modes if these modes are enabled in the decoder by the end user. Note that the encoded audio is not compressed. Instead the encoder sends DRC metadata in the bitstream to direct the behavior of the compressor in the decoder. The Dolby Media Encoder includes five compression presets designed to be appropriate for typical content. Choose the profiles that you feel sound best for your program, regardless of the profile name.

The dialogue level value determines the bitstream `dialnorm` parameter, and sets the position of the compressor null band where the signal is neither boosted nor attenuated. The null band is designed to protect audio at the average level of the program from having either cut or boost applied when dynamic range control is applied in the decoder.

[Figure 9-3](#) shows how the compression modifies the dynamic range.



**Figure 9-3** Dynamic Range Control Profile Schematic

Compression choices are available to the content producer for both Line mode and RF mode. The content producer chooses which of these profiles to assign to each mode; when the consumer or decoder selects a dynamic range control mode, the profile chosen by the producer is applied.

The DRC system metadata provides both gentle compression for graceful control of excessive dynamic range and overload protection. For example, consider a 5.1-channel program with signals at digital full scale on all channels being played through a stereo, downmixed, line-level output. Without some form of attenuation or limiting, the output signal would obviously clip. Correct setting of the dialogue level and dynamic range control profiles normally prevents clipping and unnecessary application of automatic overload protection.



**Note:** Dynamic range control profile settings are dependent on an accurate dialogue level setting. Improper setting of the dialogue level parameter may result in excessive and audible application of overload-protection limiting. Loudness measurement tools (such as the Dolby Media Meter) can reliably measure program loudness, without dialogue gating of the measured audio.

The **Film Light** setting is the default option. It provides moderate compression parameters in the bitstream that the listener may choose to apply during playback.

There are five **Late-Night Listening Mode** settings:

- Film Standard
  - Max boost: 6 dB (below -43 dB)
  - Boost range: -43 to -31 dB (2:1 ratio)
  - Null band width: 5 dB (-31 to -26 dB)
  - Early cut range: -26 to -16 dB (2:1 ratio)
  - Cut range: -16 to +4 dB (20:1 ratio)

- Film Light
  - Max boost: 6 dB (below -53 dB)
  - Boost range: -53 to -41 dB (2:1 ratio)
  - Null band width: 20 dB (-41 to -21 dB)
  - Early cut range: -21 to -11 dB (2:1 ratio)
  - Cut range: -11 to +4 dB (20:1 ratio)
- Music Standard
  - Max boost: 12 dB (below -55 dB)
  - Boost range: 55 to -31 dB (2:1 ratio)
  - Null band width: 5 dB (-31 to -26 dB)
  - Early cut range: -26 to -16 dB (2:1 ratio)
  - Cut range: -16 to +4 dB (20:1 ratio)
- Music Light (No Early Cut Range)
  - Max boost: 12 dB (below -65 dB)
  - Boost range: -65 to -41 dB (2:1 ratio)
  - Null band width: 20 dB (-41 to -21 dB)
  - Cut range: -21 to +9 dB (2:1 ratio)
- Speech
  - Max boost: 15 dB (below -50 dB)
  - Boost range: -50 to -31 dB (5:1 ratio)
  - Null band width: 5 dB (-31 to -26 dB)
  - Early cut range: -26 to -16 dB (2:1 ratio)
  - Cut range: -16 to +4 dB (20:1 ratio)

## 9.6 Timecode Subpage for Dolby Digital, Dolby Digital Plus, and Dolby TrueHD

The **Timecode** subpage in the **Encoder Setup** provides displays and options to enable or disable inclusion of SMPTE time stamps in the output bitstream, as well as to choose the timecode format and starting time for the embedded timecode (if present). You can also specify a time range, which allows you to encode just a portion of the input file, and you can choose to prepend silence to the beginning of the file in increments of frames or milliseconds.



**Note:** When you change the encode range frame-rate selection for **Encode Start** and **Encode End** (via the **Encode Start** time range drop-down menu), the start and end times will be reset to the whole frame (or millisecond) value closest to the start and end of the input file. Set the frame rate for the encoding job before entering different start and end points.

Use the <Tab> key to navigate through available entry fields. Modified values will be saved when moving the cursor to the next field using <Tab> or clicking outside the field with the mouse. Press <Enter> to update the entry in a field (and stay in the current field).

The **Timecode** subpage provides the same displays and options for Dolby Digital, Dolby Digital Plus, and Dolby TrueHD.



The **Timecode** subpage provides the following:

- **Dolby Atmos Master Offset Time** display of input timecode; Appears when encoding from a Dolby Atmos master file set with embedded timecode. The display shows the initial timecode stamp at the start of an .atmos master (in hrs:min:sec:msec).
- **Specify Time Range to Encode:** Click (check) to specify a range that is a portion of the input file(s) to encode. By default, no range is specified and the entire file is encoded.
- **Time Range Relative To** drop-down menu: This option lets you select whether the specified position is relative to the **Embedded Timecode** position or **File Position** with the start of the file treated as 00:00:00:00. The default is **Embedded Timecode**. If this option is used for files without embedded timecode, a starting timecode of 00:00:00:00 will be used.
- **Encode Start** field: Use this setting to define a valid time point from which to start encoding.
- **Encode End** field: Use this setting to define a valid time point at which to end encoding.
- Encode range drop-down menu for **Encode Start** and **Encode End** timecode: Use this setting to define the frame rate or real-time value of the encoding. The default is 23.976 fps. There are eight options.

Use one of the ms (millisecond) options if you need to select a start point that is between two video frame boundaries, as follows:

- **Real Time (ms):** Use this option if you need to edit the encode range in exact milliseconds and the Dolby Atmos file set was recorded in a “real time” SMPTE frame rate (such as 24, 25, or 30 fps). This option treats one second of the entered time as exactly one second of real time.
- **NTSC –0.1%:** Use this option if you need to edit the encode range in NTSC milliseconds and the Dolby Atmos master file set was recorded at a slower NTSC frame rate (such as 23.976 or 29.97 non-drop fps). This option treats the entered time as slower than real time, by a factor of 1.001 of real time, or –0.1%. NTSC seconds last longer than real seconds, so in samples, one second of NTSC time is 48048 samples in duration, not 48000. Refer to SMPTE ST 12-1:2014, *Time and Control Code* for further information on NTSC versus Real Time frame rates.
- **Specify New Output Timecode** drop-down menu: Use this menu if you wish to define and include timecode in the output file. Options include **Do not include Timecode**, **Use input Timecode**, and **Specify Timecode**. By default, timecode is not embedded in the encoded output file.
- **Output Timecode** display: The display is grayed out or shows the current output start timecode, dependent on the **Specify New Output Timecode** setting. If **Do not include Timecode** is selected, the field is grayed out. If **Use input Timecode** is selected, this field displays the **Encode Start** timecode from the **Specify Time Range to Encode** section. If **Specify Timecode** is selected, you can change the timecode to any positive timecode value. The default is 00:00:00:00.
- fps (frame rate) drop-down menu for **Output Timecode**. This selector is grayed out, or shows the current output frame rate, dependent on the **Specify New Output Timecode** setting. If **Do not include Timecode** is selected, the field is grayed out. If **Use input Timecode** is selected, the field uses the **Specify Time Range to Encode** fps setting, unless it is set to milliseconds, in which case the **Output Timecode** selector can be changed to any frame rate. If **Specify Timecode** is selected, you can also change the frame rate from the default of 23.976.

- **Prepend Silence** field and up/down arrows: Use the up/down arrows to set the duration of silence to add at the beginning of the encoded file. This field accepts valid integers for frames or milliseconds, based on the selected rate (in the **Prepend Silence** drop-down menu). The maximum setting is equivalent to 60 seconds.
- **Prepend Silence** drop-down menu: Use this menu to set whether the **Prepend Silence** value is based on a frame rate (23.976, 24, 25, 29.97, 29.97 df, or 30), **Real Time (ms)**, or **-0.1% NTSC (ms)**.

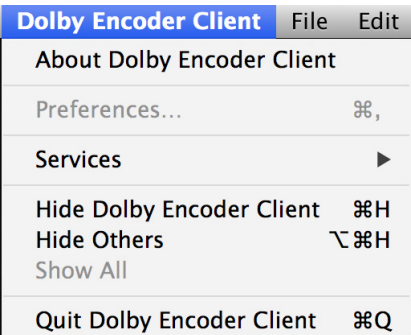
# Application Menus

These sections describe the Dolby® Media Encoder menus:

- [Dolby Encoder Client Menu](#)
- [File Menu](#)
- [Job Menu](#)
- [Window Menu](#)
- [Settings Menu](#)
- [Help Menu](#)

## 10.1 Dolby Encoder Client Menu

The **Dolby Encoder Client** menu contains standard software application menu commands.



**Figure 10-1** Dolby Encoder Client Menu

These include:

- Obtaining version information about the program
- Hiding either the application window or all other program windows
- Quitting the Dolby Media Encoder

## 10.2 File Menu

The **File** menu contains commands relating to encoder projects.

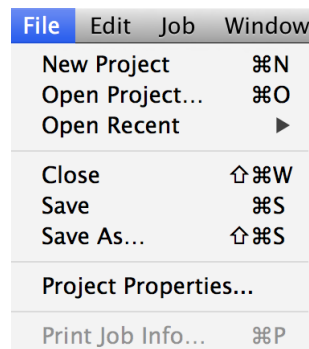


Figure 10-2 File Menu

These include:

- **New Project:** Starts a new project
- **Open Project:** Opens a stored project
- **Open Recent:** Opens a project in the system recent list
- **Close:** Closes the current project
- **Save:** Saves the current project
- **Save As:** Saves the current project with a new name
- **Project Properties:** Opens the project properties dialog
- **Print Job Info:** Opens a standard dialog allowing you to print a detailed job report (equivalent to clicking the **Print** button in the job editor)

## 10.3 Edit Menu

The **Edit** menu includes basic commands for managing text in text fields.

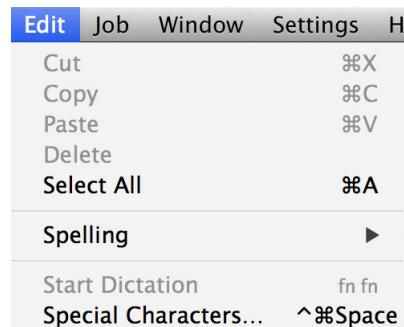


Figure 10-3 Edit Menu

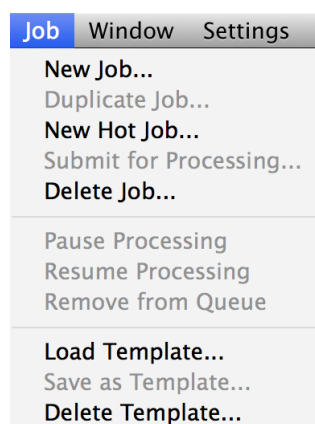
These include:

- **Cut:** Removes highlighted text in a text field, after copying it
- **Copy:** Copies highlighted text in a text field
- **Paste:** Adds the most recently copied (or cut) text into a text field (or another application)
- **Delete:** Removes highlighted text in a text field

- **Select All:** Selects all text in a text field
- **Spelling:** Provides access to menu options to check spelling and grammar:
  - **Show Spelling and Grammar:** Open the **Spelling and Grammar** dialog, where you can review each spelling and grammar issue in the current text field.
  - **Check Spelling:** highlight (underline in red) spelling and grammar issues in current text field
  - **Check Spelling While Typing:** highlight (underline in red) spelling and grammar issues as text is being entered
- **Special Characters:** Provides access to symbols and other special characters for use in text fields

## 10.4 Job Menu

The **Job** menu contains commands relating to jobs.



**Figure 10-4** Dolby Encoder Job Menu

These include:

- **New Job:** Opens the job editor dialog and starts a new job in the current project.
- **Duplicate Job:** Creates an exact duplicate of the selected job with the suffix *\_copy*. All fields except the destination are copied. An entry saying *Job Duplicate from Xxxxxx* is added to the job log. This command makes it easy to redo a job that has already been encoded and is therefore locked.
- **New Hot Job:** Opens the job editor dialog and creates a new Hot Job.
- **Submit for Processing:** Submits the job for encoding; equivalent to clicking the **Submit** button in the job editor.
- **Delete Job:** Removes the selected job; equivalent to clicking the **Remove** button in the job list window.
- **Pause Processing:** Pauses processing; equivalent to clicking the **Pause** button in the job list window (when the encoder queue is displayed).
- **Resume Processing:** Resumes processing; equivalent to clicking the **Resume** button in the job list window (when the encoder queue is displayed).
- **Remove from Queue:** Removes the current job from the server queue; equivalent to clicking the **Remove** button in the job list window (when the encoder queue is displayed).

- **Load Template:** Opens a standard **Open** dialog, letting you choose an existing .djt template. Loading a template creates a new job in the current project and launches a new job editor using the settings of the template.
- **Save as Template:** Opens a standard **Save As** dialog, letting you save the currently opened job as a template for future use. A template file has the .djt extension.
- **Delete Template:** Opens a standard delete dialog, allowing you to choose a template to remove.



**Note:** The job-related commands **Move Up** and **Move Down** are available in the encoder queue in the job list window only.

## 10.5 Window Menu

The **Window** menu contains commands that open and close the job list window, as well as any active job editor windows.

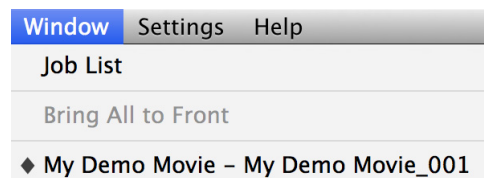


Figure 10-5 Window Menu

## 10.6 Settings Menu

The **Settings** menu contains commands that provide access to controls for configuring the system.

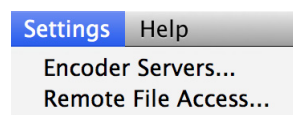


Figure 10-6 Dolby Encoder Settings Menu

These include:

- **Encoder Servers**
- **Remote File Access**

## 10.7 Help Menu

The **Help** menu provides access to a .pdf file of the *Dolby Media Encoder User's Manual*.

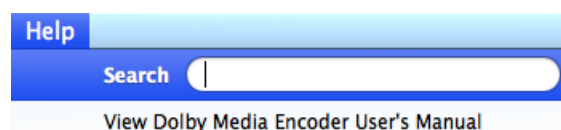


Figure 10-7 Help Menu

## Reference Information

This chapter provides the following reference information:

- File-name conventions that ensure automatic file selection. See [Section 11.1](#).
- Dolby® professional technologies. See [Section 11.3](#).

For reference information on encoding options for Dolby TrueHD, Dolby Digital, Dolby Digital Plus™, and MLP Lossless™ formats, see [Chapter 8](#).

### 11.1 File-Name Conventions that Ensure Automatic File Selection

The system will automatically select source files if the file names follow these rules:

1. All of the channel IDs used are listed in [Table 11-1](#). The channel IDs are not case sensitive.
2. The channel ID is separated before and after with either a period or an underscore.
3. The file names for the channels are identical except for channel ID.
4. The number of files matches the number of channels selected in the encoder **Channels** page.
5. The file channel IDs match the channel IDs selected in the client **Channels** page.
6. File names do not contain the ampersand (&) character.



**Note:** File names including the ampersand (&) character cannot be processed by Dolby Media Producer.

[Table 11-1](#) is adapted from SMPTE 428M to list the channel IDs supported in the Dolby Media Producer software. IDs are not case sensitive.

**Table 11-1** Supported Channel Labels with Descriptions

| Label        | Name                  | Location                                            |
|--------------|-----------------------|-----------------------------------------------------|
| L, Lt        | Left                  | Far-left screen loudspeaker                         |
| R, Rt        | Right                 | Far-right screen loudspeaker                        |
| C            | Center                | Center screen loudspeaker                           |
| LFE, LF, SW  | Low-Frequency Effects | Screen Low-Frequency Effects subwoofer loudspeakers |
| Ls, Ssl, Lss | Left Surround         | Left-wall surround loudspeakers                     |
| Rs, Ssr, Rss | Right Surround        | Right-wall surround loudspeakers                    |
| Lc           | Left Center           | Mid-left to center-screen loudspeaker               |

**Table 11-1** Supported Channel Labels with Descriptions

| Label                 | Name                   | Location                                                       |
|-----------------------|------------------------|----------------------------------------------------------------|
| Rc                    | Right Center           | Mid-right to center-screen loudspeaker                         |
| Lrs, Rls,<br>Bsl, Lsr | Left Rear Surround     | Rear-wall left loudspeakers                                    |
| Rrs, Bsr,<br>Rsr      | Right Rear Surround    | Rear-wall right loudspeakers                                   |
| Cs                    | Center Surround        | Rear-wall surround loudspeakers                                |
| Ts, Oh                | Top Surround           | Center-of-the-theatre ceiling loudspeakers                     |
| Lsd                   | Left Surround Direct   | Left-surround single loudspeaker for localized directionality  |
| Rsd                   | Right Surround Direct  | Right-surround single loudspeaker for localized directionality |
| Lw                    | Left Wide              | Outside the screen, front left loudspeaker                     |
| Rw                    | Right Wide             | Outside the screen, front right loudspeaker                    |
| Vhl, Lh               | Vertical Height Left   | Far-left top-of-screen loudspeaker                             |
| Vhc, Ch               | Vertical Height Center | Center top-of-screen loudspeaker                               |
| Vhr, Rh               | Vertical Height Right  | Far-right top-of-screen loudspeaker                            |
| LFE2                  | Secondary LFE          | Secondary Low-Frequency Effects subwoofer loudspeaker          |

The following sections provide four examples of file sets that would be automatically selected.

### 11.1.1 Example 1

This file set uses a period as the separator:

- ExampleOne.L.wav
- ExampleOne.R.wav
- ExampleOne.C.wav
- ExampleOne.LFE.wav
- ExampleOne.Ls.wav
- ExampleOne.Rs.wav

### 11.1.2 Example 2

This file set uses an underscore as the separator:

- ExampleTwo\_L.wav
- ExampleTwo\_R.wav
- ExampleTwo\_C.wav
- ExampleTwo\_LFE.wav
- ExampleTwo\_Ls.wav
- ExampleTwo\_Rs.wav



### 11.1.3 Example 3

In this file set, the channel ID is not the last element in the file name. Again, the file set uses a period as the separator:

- ExampleThree.L.ENG.wav
- ExampleThree.R.ENG.wav
- ExampleThree.C.ENG.wav
- ExampleThree.LF.ENG.wav
- ExampleThree.LS.ENG.wav
- ExampleThree.RS.ENG.wav

### 11.1.4 Example 4

In this file set, the channel ID is not the last element in the file name. The file set uses an underscore as the separator:

- ExampleFour\_L\_ENG.wav
- ExampleFour\_R\_ENG.wav
- ExampleFour\_C\_ENG.wav
- ExampleFour\_SW\_ENG.wav
- ExampleFour\_LS\_ENG.wav
- ExampleFour\_RS\_ENG.wav

## 11.2 Downmixing in Dolby Systems

Stereo and 7.1 to 5.1 downmixing settings in the Dolby Media Encoder are based on the encoding format, target, and expected listening environments.

### Stereo Downmixing

You set stereo downmixing for Dolby TrueHD presentations and Dolby Digital Plus and Dolby Digital channel-based material in the **Downmix** subpage of the **Encoder Setup**.

For Dolby TrueHD, the **6 to 2 Channel Downmix** setting defines the stereo downmix. For Dolby Digital Plus and Dolby Digital, the **Preferred Stereo Downmix** setting will set the metadata for the stereo downmix.

The following stereo downmix options are supported:

- Dolby TrueHD
  - Standard: Two-channel stereo Left, Right (Lo/Ro)
  - Custom: Includes adjustable controls for setting levels
- Dolby Digital Plus and Dolby Digital
  - Surround: Two-channel Left, Right (a matrix-encoded Lt/Rt signal)
  - Stereo: Two-channel stereo Left, Right (Lo/Ro)
  - Dolby Pro Logic II® (Dolby Digital Plus and online media only): Two-channel Left, Right (a Dolby Pro Logic II matrix-encoded Lt/Rt signal)

For Dolby Digital Plus and Dolby Digital, phase 90 is applied by enabling **Surround Channel 90-Degree Phase Shift** in the **Preprocessing** subpage of the **Encoder Setup**. This option is enabled by default.



**Note:** The phase 90 filters used by default provide the all-pass 90-degree phase shift filtering for the Ls/Rs feeds into the downmix. This reduces undesirable signal cancellation in stereo presentations, improves imaging, and enables proper matrix decoding. It is strongly recommended to use the 90-degree phase shift for any Lt/Rt downmixes.

All the stereo downmix options are created by first creating a 5.1 downmix from the 7.1 content, and then downmixing the 5.1 to 2.0 in the same manner as 2.0 consumer products.

The coefficients for the two-channel downmixes from 5.1 are as follows:

Lt/Rt (legacy):

$$Lt = L + (-3 \text{ dB} \times C) - (-3 \text{ dB} \times Ls) - (-3 \text{ dB} \times Rs)$$

$$Rt = R + (-3 \text{ dB} \times C) + (-3 \text{ dB} \times Ls) + (-3 \text{ dB} \times Rs)$$

Lt/Rt (Pro Logic II):

$$Lt = L + (-3 \text{ dB} \times C) - (-1.2 \text{ dB} \times Ls) - (-6.2 \text{ dB} \times Rs)$$

$$Rt = R + (-3 \text{ dB} \times C) + (-6.2 \text{ dB} \times Ls) + (-1.2 \text{ dB} \times Rs)$$

LoRo:

$$Lo = L + (-3 \text{ dB} \times C) + (-3 \text{ dB} \times Ls)$$

$$Ro = R + (-3 \text{ dB} \times C) + (-3 \text{ dB} \times Rs)$$

## 7.1 to 5.1 Downmixing

Dolby Media Encoder creates signals that carry 5.1 and 7.1 presentations, as well as object audio for rendering. The type of downmix to 5.1 is selected when encoding.

For Dolby TrueHD, the **8- to 6-Channel Downmix** setting (in the **Downmix** subpage of the **Encoder Setup**) defines 5.1 downmixing. For Dolby Digital Plus and Dolby Digital, the job editor window provides controls for defining 5.1 downmixing (**5.1 Core Type** for Dolby Digital Plus, **5.1 Downmix Type** for Dolby Digital).

The following downmix options are supported:

- Dolby TrueHD
  - Dolby Pro Logic IIx: Dolby Pro Logic IIx matrix-encoded 5.1 mix, with the equation coefficients noted below
  - Standard (Lo, Ro): 5.1 Standard mix, with the equation coefficients noted below
  - Custom: Includes adjustable controls for setting levels
- Dolby Digital Plus and Dolby Digital
  - Dolby Pro Logic IIx (Dolby Digital Plus and online media only): Dolby Pro Logic IIx matrix-encoded 5.1 mix, with the equation coefficients noted below
  - 5.1 Standard (Lo, Ro): 5.1 Standard mix, with the equation coefficients noted below.

The coefficients for a Dolby Pro Logic PLIIx downmix from 7.1 to 5.1 are:

$$Ls = Lss + (-1.2 \text{ dB} \times Lrs) + (-6.2 \text{ dB} \times Rrs)$$

$$Rs = Rss + (-6.2 \text{ dB} \times Lrs) + (-1.2 \text{ dB} \times Rrs)$$

The coefficients for a 5.1 Standard (Lo, Ro) downmix from 7.1 to 5.1 are:

$$L_s = 0 \text{ dB} \times L_{ss} + 0 \text{ dB} \times L_{rs}$$
$$R_s = 0 \text{ dB} \times R_{ss} + 0 \text{ dB} \times R_{rs}$$

## 11.3 Dolby Professional Technologies

The Dolby Media Encoder supports all of the following Dolby audio technologies:

- Dolby Atmos®
- Dolby TrueHD
- Dolby TrueHD
- Dolby Digital
- MLP Lossless
- Dolby Surround Pro Logic
- Dolby Pro Logic II
- Dolby Pro Logic IIx
- Dolby Pro Logic IIz
- Dolby Digital Surround EX™
- Dolby Headphone™

## 11.4 Dolby Atmos

With Dolby Atmos, sound comes alive from all directions, including overhead. The Dolby Atmos format combines a channel-based audio bed with object-oriented sound to place and move specific effects around the listener. Using dynamic audio objects, sound designers and artists are free to mix audio in a three-dimensional space, steering effects throughout the room. When the Dolby Atmos soundtrack is played back in the cinema, the object-based audio mix is adapted to the room. (That is, the audio is rendered to the exact speaker layout of the theatre.)

A major benefit of Dolby Atmos is the ability to scale a soundtrack to different room sizes and speaker configurations. Dolby Atmos provides an optimal presentation of the soundtrack for each listening environment.

Audio/video receiver and home-theater-in-a-box products were the first consumer products to support Dolby Atmos. These products decode, scale, and calibrate Dolby Atmos content to accommodate a variety of home theater configurations. The initial Dolby Atmos products can support as few as 7 speakers and as many as 34 speakers.

Dolby Atmos for home theater has the following features to produce an incredible, enveloping, three-dimensional audio experience:

- Object-based Dolby Atmos content
- Dolby Atmos decoders
- Dolby advanced audio postprocessing
- Dolby Atmos enabled speaker technology

## 11.5 Dolby TrueHD

Dolby TrueHD is Dolby's lossless technology for high-definition disc-based media. Dolby TrueHD delivers sound that is bit-for-bit identical to the studio master, unlocking the true high-definition entertainment experience on current and future surround formats on Blu-ray Disc. When coupled with high-definition video, Dolby TrueHD offers an unprecedented home theater experience that lets you enjoy sound as stunning as the high-definition picture.

### 11.5.1 Features

The most notable features of Dolby TrueHD are:

- Lossless coding technology
- Supports carriage of Dolby Atmos content
- Supports data rates up to 18 Mbps
- Supports up to eight full-range channels of 24-bit/96 kHz audio and up to six channels of 24-bit/192 kHz audio
- Supported by HDMI™
- Supports extensive metadata, including dialogue normalization and dynamic range control
- Supports multiple discrete presentations that can be custom mixes for Dolby Atmos, 7.1, 5.1, or stereo

Dolby TrueHD can support more than eight audio channels, although Blu-ray Disc standards currently limit the maximum number of audio channels to eight.

### 11.5.2 Benefits

The most notable benefits of Dolby TrueHD are:

- Studio master-quality sound that unlocks the true high-definition entertainment experience on next-generation discs.
- Compatible with the audio/video (A/V) receivers and home-theater-in-a-box products of today and tomorrow.
- Dialogue normalization that can maintain the same volume level when you change to other Dolby True HD, Dolby Digital Plus, and Dolby Digital programming.
- Dynamic range control (Night mode) enables you to customize audio playback to reduce peak volume levels while experiencing all the details in the soundtrack.
- Selected as a format for Blu-ray Disc™.

### 11.5.3 Compatibility with Current A/V Receivers

Whether you own an older A/V system equipped with Dolby Digital, or a newer A/V receiver equipped with an HDMI connection or external line-level multichannel inputs, you are ensured a compatible, high-quality playback path.

## 11.6 Dolby Digital Plus

Dolby Digital Plus is the next-generation audio technology for all high-definition programming and media. It combines the efficiency to meet future broadcast demands with the power and flexibility to realize the full audio potential of the high-definition experience. Built on Dolby Digital, the multichannel audio standard for DVD and HD broadcasts worldwide, Dolby Digital Plus was designed for the delivery formats of the future, but remains fully compatible with all current A/V receivers. With Dolby Digital Plus, you get even higher quality audio, more channels, and greater flexibility.

### 11.6.1 Features

The most notable features of Dolby Digital Plus are:

- Multichannel sound with discrete channel output.
- Supports carriage of Dolby Atmos
- Channel and program extensions can carry multichannel audio programs of up to 7.1 channels and support multiple programs in a single encoded bitstream.
- Outputs a Dolby Digital bitstream for playback on existing Dolby Digital systems.
- Supports data rates as high as 6 Mbps.
- Bit-rate performance of up to 1.6 Mbps on Blu-ray Disc.
- Accurately reproduces what the director and producer intended.
- Interactive mixing and streaming capability in advanced systems.
- Supported by High-Definition Media Interface™ (HDMI).

### 11.6.2 Benefits

The most notable benefits of Dolby Digital Plus are:

- Can deliver 7.1 channels or more enhanced-quality audio at up to 6 Mbps.
- Allows multiple languages to be carried in a single bitstream.
- Offers audio professionals new creative power and freedom.
- Compatible with the millions of home entertainment systems equipped with Dolby Digital.
- No latency or loss of quality in the conversion process. Maintains high quality at more efficient broadcast bit rates (<192 kbps for 5.1-channel audio).
- Selected by the Advanced Television Systems Committee (ATSC) as the standard for future broadcast applications; named as an option by the Digital Video Broadcasting (DVB®) Project for satellite and cable TV.
- Selected as a format for Blu-ray Disc.

Next-generation high-definition disc players are compatible with current audio/video (A/V) receivers. Whether you own an older A/V system equipped with Dolby Digital, or a newer A/V receiver equipped with an HDMI connection or external line-level multichannel inputs, you are ensured a compatible, high-quality playback path.

Dolby Digital Plus can support more than eight audio channels, even though Blu-ray Disc standards currently limit their maximum number of audio channels to eight.

## 11.7 Dolby Digital

In thousands of cinemas and millions of homes worldwide, Dolby Digital is the standard for surround sound technology in general and 5.1-channel surround sound in particular.

Dolby Digital is a highly sophisticated and versatile audio encoding/decoding technology. Dolby Digital technology can transmit mono, stereo (two channel), or up to 5.1-channel surround sound (discrete multichannel audio).

The superior coding efficiency of Dolby Digital—and its ability to deliver high-quality discrete multichannel audio without compromising video quality—has made it the designated audio standard for DVD worldwide.

Dolby Digital is also the preferred multichannel audio standard for broadcast direct to home satellite and digital cable systems. Nearly 60 million digital cable and satellite set-top boxes are currently in use worldwide, supported by over 30 million audio/video receivers equipped with 5.1-channel Dolby Digital decoding. Virtually every DVD-Video player sold today offers a two-channel downmix of the Dolby Digital 5.1-channel movie soundtrack, as well as a digital output for connection to a 5.1-channel audio/video receiver.

In the film industry, the Dolby Digital soundtrack is optically encoded right on the filmstrip, in the space between the sprocket holes. Placing the soundtrack directly on the film allows it to coexist with the analog track, a cost-effective and simple solution for film distributors and theatre owners. The sprocket hole area has also proven highly resistant to wear and tear, so that a soundtrack encoded in Dolby Digital will remain free of pops and hiss for the useful life of the print.



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**Note:** The sound information contained in each of the six available channels is distinct and independent. These six channels are described as a 5.1-channel system, because there are five full-bandwidth channels (Left, Right, Center, Left Surround, and Right Surround) with a 3 Hz to 20 kHz frequency range, plus a Low-Frequency Effects (LFE) subwoofer channel devoted to frequencies from 3 to 120 Hz.

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## 11.8 MLP Lossless

MLP Lossless, licensed by Dolby Laboratories, is the core technology of Advanced Resolution multichannel and stereo DVD-Audio.

MLP Lossless enables producers to encode up to six channels of 24-bit/96 kHz audio, or two channels of 24-bit/192 kHz audio onto a DVD-Audio disc, resulting in playback that is bit-for-bit identical to the studio master. Nothing is lost during the encoding and decoding process.

Widely recognized as the first major step forward in high-fidelity audio after the compact disc, DVD-Audio makes your home audio system sound like a live musical performance. DVD-Audio combines DVD technology with MLP Lossless coding to give you music with full 5.1-channel surround sound and fidelity that is dramatically better than that of the CD. A DVD-Audio player provides an excellent listening experience for DVD-Audio discs, but many DVD-Video player can also play DVD-Audio discs.

## 11.9 Dolby Surround Pro Logic

Dolby Surround is the consumer version of the original Dolby multichannel analog film sound formats (Dolby Stereo and Dolby SR). When a Dolby Surround soundtrack is produced, four channels of audio information—Left, Center, Right, and mono Surround—are matrix encoded onto two audio tracks. These two tracks are then carried on stereo program sources such as stereo soundtracks on DVDs and Blu-ray Discs, videotapes and TV broadcasts into the home, where they can be decoded by Dolby Pro Logic to re-create the original multi-channel surround sound experience.

A result of Dolby's expertise in matrix surround processing, Dolby Pro Logic was the foundation for multichannel home theater, and was the reference decoder for creating the surround sound audio tracks in thousands of commercially available videocassettes, laser discs, DVDs, and television programs.

With the introduction of the Dolby Digital multichannel film sound format, Dolby Digital replaced Dolby Surround as the preferred technology to deliver multichannel audio to consumers via DVD-Video, digital television, and games. However, every Dolby Digital decoder also provides a stereo signal compatible with Dolby Pro Logic on its analog outputs. See [Dolby Digital](#) for additional details.

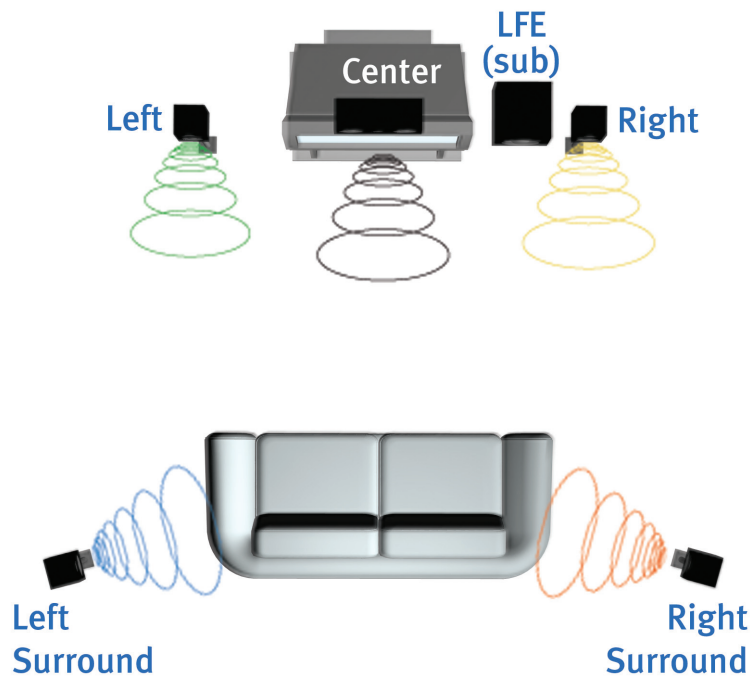
## 11.10 Dolby Pro Logic II

Dolby Pro Logic II technology processes any high-quality stereo (two-channel) audio into five playback channels of full-bandwidth surround sound. A matrix surround decoding technology, Dolby Pro Logic II detects the directional cues that occur naturally in stereo content and uses these elements to create a five-channel surround sound playback experience. It is ideally suited for home theater systems, PCs, game consoles, and multichannel in-car audio systems.

Dolby Pro Logic II is fully compatible with all Dolby Pro Logic technologies. It provides optimal audio for playback in a 5.1-channel home theater system for the thousands of videocassettes and TV programs encoded in four-channel Dolby Surround (the encoding counterpart to the Dolby Pro Logic decoding technology). Dolby Pro Logic II also enables video game consoles to encode five-channel surround sound information into a stereo signal with virtually no impact on the console CPU, which means better audio with no impact on game play.

Dolby Pro Logic II can bring new life to your CD collection. Play your music through your home theater system, and you will hear seamless, natural sound with new depth and detail—almost like hearing a newly remastered CD.

In addition to enhancing playback, Dolby Pro Logic II can be used to encode TV programming to deliver a surround sound experience for viewers with stereo TV systems.



**Figure 11-1** Dolby Pro Logic II Speaker Configuration

## 11.11 Dolby Pro Logic IIx

Dolby Pro Logic IIx was designed to expand upon several multichannel technologies, including Dolby Pro Logic II matrix decoding as well as Dolby Digital and Dolby Digital EX audio systems, by deriving an additional pair of independent rear surround channels. This additional channel pair offers decoders additional flexibility in deriving up to 7.1 channels to feed all available speaker outputs of a surround sound system, and to do so from a wide range of sources (from stereo to discrete multichannel). While 7.1 rear surround channels add an extra dimension to surround sound systems, all outputs of Dolby Pro Logic IIx are currently intended for speaker locations fixed on a horizontal plane.

### 11.11.1 Features

The most notable features of Dolby Pro Logic IIx are:

- Ability to generate up to 7.1 channels from stereo and 5.1 sources
- Support for the following decoding modes:
  - Music mode
  - Movie mode
  - Matrix mode
  - Dolby Digital EX mode



- Support for the following upmixed output configurations based on stereo sources:
  - 2 to 3 (L and R to L, C, and R)
  - 2 to 4 (L and R to L, R, Ls, and Rs)
  - 2 to 5.1 (L and R to L, C, R, Ls, and Rs)
  - 2 to 6.1 (L and R to L, C, R, Ls, Rs, and Cs)
  - 2 to 7.1 (L and R to L, C, R, Ls, Rs, Lrs, and Rrs)
- Support for the following upmixed output configurations based on 5.1 sources with Independent Ls and Rs channels:
  - 5.1 to 6.1 (Ls and Rs to Ls, Rs, and Cs)
  - 5.1 to 7.1 (Ls and Rs to Ls, Rs, Lrs, and Rrs)



**Note:** For 5.1-channel sources, Dolby Pro Logic IIx operates on the Ls and Rs channels only; the L, C, R, and LFE channels are bypassed.

## 11.12 Dolby Pro Logic IIz

Dolby Pro Logic IIz is an extension to Dolby Pro Logic IIx. It was developed to improve the spatial experience offered by existing surround sound systems by offering the ability to derive up to 9.1 channels that include height outputs.



**Note:** The Dolby Media Decoder currently supports up to eight output channels and will support the Pro Logic IIz 7.1-with-Height modes.

### 11.12.1 Features

Dolby Pro Logic IIz includes all of the features of Dolby Pro Logic IIx, in addition to the following notable features:

- Ability to generate audio of up to 9.1 channels that includes height channel outputs from stereo, 5.1, and 7.1 Back sources
- Supports the following upmixed output configurations based on stereo sources:
  - 2 to 7.1 Height (L and R to L, C, R, Ls, Rs, Lvh, and Rvh)
  - 2 to 9.1 (L and R to L, C, R, Ls, Rs, Lrs, Rrs, Lvh, and Rvh)
- Supports the following upmixed output configurations based on 5.1 sources with independent Ls and Rs channels:
  - 5.1 to 7.1 Height (Ls and Rs to Ls, Rs, Lvh, and Rvh)
  - 5.1 to 9.1 (Ls and Rs to Ls, Rs, Lrs, Rrs, Lvh, and Rvh)



**Note:** For 5.1-channel sources, Dolby Pro Logic IIz operates only on the Ls and Rs channels; L, C, R, and LFE channels are bypassed.

Because high-definition formats allow content to contain up to 7.1 discrete channels from Dolby Digital Plus and Dolby TrueHD audio programs, Dolby Pro Logic IIz also supports 7.1 Back sources and offers the following upmixed output configurations:

- 7.1 Back to 7.1 Height (Ls, Rs, Lrs, Rrs to Ls, Rs, Lvh, and Rvh)
- 7.1 Back to 9.1 (Ls and Rs to Ls, Rs, Lvh, and Rvh)



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**Note:** For 7.1-channel sources, Dolby Pro Logic IIz may operate only on the Ls, Rs, Lrs, and Rrs channels; L, C, R, and LFE are bypassed.

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While Dolby Pro Logic IIz can be used to upmix a variety of sources such as movies and music, it is particularly suited for upmixing game content and will more effectively use existing program material when height channel/speaker outputs are present.

## 11.13 Dolby Digital Surround EX

This technology was first introduced by Dolby in the cinema, to augment the standard 5.1-channel cinematic experience with a new Back Surround channel, reproduced by speakers placed directly behind the audience. The resulting 6.1-channel format added a new dimension to the Dolby Digital 5.1 cinematic experience, and was soon introduced in DVD releases and 6.1 home theater playback equipment as well. The additional channel was matrix-encoded onto the regular Left Surround and Right Surround channels of a conventional Dolby Digital 5.1 soundtrack. This allowed it to be either matrix-decoded by a Dolby Digital Surround EX system, or reproduced by the traditional Left Surround and Right Surround speakers in an older 5.1 system. An EX flag in the bitstreams allowed home theater equipment to detect content with this matrix-encoded Back Surround channel and reproduce it correctly.

The introduction of Dolby Pro Logic IIx 7.1 channel home theater systems surpassed the 6.1 experience and gained rapid adoption. As described [Section 11.11](#), the PLIIx system matrix encodes not one but two rear surround signals into the conventional Ls and Rs channels. The EX flag in the bitstream is re-purposed to indicate presence of any matrix-encoded rear surround signals in a Dolby TrueHD, Dolby Digital Plus or Dolby Digital 5.1-channel presentation, including the 5.1-channel presentation heard when decoding a 7.1-channel bitstream on a system with only 5.1 capability.

## 11.14 Dolby Headphone

Dolby Headphone technology allows users to wear any set of headphones and listen to music, watch movies, or play video games with the dramatic surround effects of a 5.1-channel soundtrack. Headphone listeners can enjoy the spacious, natural soundfield for hours.

### 11.14.1 Features

The most notable features of Dolby Headphone are:

- Realistic 5.1-channel surround sound from any set of headphones
- Processes 24-bit/96 kHz audio, such as DVD-Audio, to produce surround sound bit-for-bit identical to the studio masters
- Superb surround sound with Dolby Pro Logic II from stereo music and movies

### 11.14.2 Benefits

The most notable benefits of Dolby Headphone are:

- Enables enjoyment of surround sound without disturbing others
- Improves listening comfort and reduces listener fatigue
- Allows stereo content to be heard as 5.1-channel surround sound (when combined with Dolby Pro Logic II)

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## Preparing Secondary Audio and Automation in Pro Tools

Mixing in Avid® Pro Tools® for secondary audio is essentially the same as a normal mix. The difference is, instead of exporting a finished audio mix, you will export the mix automation data that you create for your primary audio track and embed that data into your Dolby® Digital Plus™ secondary audio stream. The Blu-ray Disc™ player will mix your secondary audio against any primary track on your disc using this data. This allows for a great savings in data rate and data storage when authoring your Blu-ray Disc. It also allows the secondary track to be delivered via broadband to an Internet-capable player after the disc has been released. Because your secondary audio will be mixed downstream, you must take care to ensure your secondary audio source files and mix automation track will mix as you intended.

It is best if the primary track you plan to mix to is the exact same track that is on your Blu-ray Disc. Ensure that this file has the correct start time, and was not trimmed during authoring. An incorrect start time could result in your mix metadata being offset to your track.

It is best if your secondary audio track is mixed and final before you begin encoding. If you do make changes to your secondary track in the mix session, be sure to export the audio tracks with all of your changes to use in the Dolby Media Encoder.

If you plan to use a single automation track for all channels, it is best to import your primary track as an interleaved file. If you are mixing with different levels for different channels, you can use either interleaved or multiple mono files. Remember, a single automation track can have up to 50 dB of attenuation, while attenuation is limited to 28 dB if channels are attenuated separately.

Read the following important information regarding your secondary audio mix.

### 12.1 Important Audio Preparation Information

Remember that the secondary audio track you create can be mixed with any primary audio track on a Blu-ray Disc. These primary tracks may have been encoded using any audio codec that supports Blu-ray Disc (as well as linear PCM). In addition, these tracks may not necessarily have the same relative level, or may have dialogue level metadata applied.



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**Note:** It is important that you are aware of all the primary audio sources that will be available on your disc.

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If the primary audio tracks on your disc have been encoded with a dialogue level value other than -31, you need to compensate for that level scaling in your mix session monitoring to ensure that you are hearing a correct balance between your primary and secondary audio. Use one of the following methods:

- If you plan to set the dialogue level value in your secondary audio encode in the same way as your primary audio was set, this will give you matching levels in your Pro Tools mix session and in the Blu-ray Disc player mixer. No action is required in your Pro Tools mix session. If your primary audio is PCM, a value of -31 in your secondary audio would be the same as matching `dialognorm` in the secondary audio stream.
- If you plan to set the secondary audio dialogue level to a different value than your primary, you will need to do some scaling of the audio in your mix session for monitoring purposes only.

## 12.2 Examples

If your primary soundtrack was encoded with a dialogue level value of -27 and your secondary audio will be encoded with a value of -31, the decoder in the Blu-ray Disc player will scale down the primary 4 dB to output the audio at -31. This would, in effect, make your primary audio 4 dB lower relative to your secondary. To account for this in your mixing session, you will need to scale down your primary audio by 4 dB for monitoring purposes only. This can easily be done in Pro Tools using the trim insert.

Conversely, if your primary soundtrack was encoded with a dialogue level value of -31 and your secondary audio will be encoded with a value of -25, the decoder in the Blu-ray Disc player will scale down the secondary 6 dB to output the audio at -31. This would, in effect, make your primary audio 6 dB higher relative to your secondary. To account for this in your mixing session, you will need to scale down your secondary audio by 6 dB for monitoring purposes only.

Be sure to remove any scaling in your mix that was applied to account for the application of dialogue level metadata in a player. Do this before exporting your .aaf file or secondary audio source files. This is to ensure that there is no extra attenuation in your exported mix automation data or secondary audio source files.

## 12.3 Exporting .aaf Files from Pro Tools

Once you have completed your mix, use these steps to properly export an .aaf file from Pro Tools:

1. Select the channels for which you wish to export metadata.
2. Choose **File > Export > Selected Tracks as OMF/AAF**.
3. From the **Export As** drop-down menu, select **AAF**.
4. Deselect **Enforce Avid Compatibility**.
5. Set the **Copy Option** to **Link to Source Media**.
6. Set other export options, as needed.
7. Click **OK**.